Roots of d-matching polynomials and the spectrum of the universal cover: a closed form at first Betti number one, a conjecture, and the biregular case

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Abstract

Hall, Puder and Sawin place the roots of the d-matching polynomial $\mu_{d,G}$ in the interval $[-\rho, \rho]$, ρ the spectral radius of the universal cover T. We conjecture that they lie in spec(T) itself, which is a proper subset of that interval whenever T has a spectral gap: no root ever enters a gap. This strengthens their theorem, is the matching-polynomial form of their Question 6.3, and contains Problem 1 of Song, Fan and Miao — its biregular case at $d = 1$ — extending it to all graphs and all d.

We prove the conjecture for every connected G of first Betti number one and every d, by giving $\mu_{d,G}$ in closed form: it is V_d for the three-term recurrence $V_d = \mu_G V_{d-1} - \mu^2_{G-V(C)} V_{d-2}$, so its roots are exactly the solutions of $\mu_G = 2\cos\big(k\pi/(d+1)\big)\mu_{G-V(C)}$ and equidistribute on spec(T). This answers Question 6.5 of Hall–Puder–Sawin for that family, and the hypothesis cannot be relaxed. For the cycle it specializes to $\Phi_{n,r} = \chi_{C_n} \cdot U_{r-1}(T_n(x/2))$, equivalent to Hall's conjecture, by an elementary route through the exponential formula. We also prove the conjecture for every subdivision at $d = 1$, which settles the case $\min(d, r) = 2$ of Problem 1 of Song, Fan and Miao.

For (a, b)-biregular G the conjecture becomes a two-sided Heilmann–Lieb statement, of which the upper half is Godsil's. We determine the top five coefficients of the recentred polynomial: four are universal in (p, a, b), and the fifth is universal plus the number of 4-cycles, with coefficient one. In operator form μ_G is the mixed characteristic polynomial of rank-b projections summing to aI, and commuting such families are *exactly* the biregular incidence families, so the projection statement is the precise noncommutative extension. A determinantal representation gives the unconditional bound $\lambda \leq ab$, which overshoots the band edge by exactly $(\sqrt{(a-1)(b-1)} - 1)^2$ — hence proves the band outright at $a = b = 2$ — and improves on Marchenko–Pastur whenever $1/\sqrt{a} + 1/\sqrt{b} > 1$. The band is the support of a free convolution *power*, $\chi^{\boxplus a}$; every method that falls short computes a compound free Poisson instead, whose free cumulants agree with the correct ones through order three and split at order four, the same order at which the coefficient ladder first sees the graph. Letting the common rank vary with the trace and the sum held fixed, the extreme roots move monotonically from the tree band at rank b to Marchenko–Pastur at rank p: each obstruction is sharp for the class it sees, and the conjecture is the assertion that minimal rank contracts the interval all the way. We also give a vertex recursion for

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weighted 2-plane families valid in every direction, not only along coordinate axes, whose cavity term is a sum of squares level by level and which is the Heilmann–Lieb recursion in the diagonal case; it reduces the band $[a-2\sqrt{a},\ a+2\sqrt{a}]$ for the whole class to the sign of a single cross term, and shows that $2\sqrt{a}$ is the fixed point of the induction rather than an artefact of the estimate. The closed form, the exponential formula, the subdivision bound, the coefficient identities including the 4-cycle term, the reflection, the ab bound and the positivity behind the extremal computation are machine-checked in Lean 4 over Mathlib, with no axioms beyond Lean's standard three.

1 Introduction and statement

The expected characteristic polynomial of a random lift is the object at the heart of the interlacing-families method of Marcus–Spielman–Srivastava [26], who used the $r = 2$ (signing) case to prove that bipartite Ramanujan graphs exist in every degree; Hall–Puder–Sawin [4] extended the framework to arbitrary coverings. We evaluate it in closed form for the cycle.

A *permutation r-lift* of a graph G replaces each vertex by r copies and each edge $\{u, v\}$ by the perfect matching $u_i \sim v_{\sigma(i)}$ of a permutation $\sigma \in S_r$ chosen independently and uniformly per edge. For the cycle C_n, gauge-fixing a spanning path reduces the lift to a single permutation $\sigma \in S_r$ on the closing edge, and the lift decomposes into disjoint cycles: each ℓ-cycle of σ contributes a cycle $C_{n\ell}$ (fixed points contribute copies of C_n). Since $\chi_{C_m}(x) = \det(xI - A_{C_m}) = 2T_m(x/2) - 2$, the expected characteristic polynomial is the expected cycle weight

$$\Phi_{n,r}(x) \;=\; \mathbb{E}_{\sigma\in S_r} \prod_{\ell\,\in\,\mathrm{cycles}(\sigma)} \big(2T_{n\ell}(x/2) - 2\big). \tag{1}$$

Theorem 1. *For all $n \geq 1$, $r \geq 1$, with $Y = T_n(x/2)$,*

$$\Phi_{n,r}(x) \;=\; U_r(Y) - 2U_{r-1}(Y) + U_{r-2}(Y) \;=\; \underbrace{\big(2T_n(x/2) - 2\big)}_{\chi_{C_n}(x)} \cdot \underbrace{U_{r-1}\big(T_n(x/2)\big)}_{\Psi_r(x)}.$$

The factored form identifies the quotient Ψ_r — by the theorem of Hall–Puder–Sawin [4] equating the expected new-eigenvalue characteristic polynomial with the d-matching polynomial ($d = r - 1$) — as

$$\mu_{d,C_n}(x) \;=\; U_d\big(T_n(x/2)\big),$$

which is equivalent (through $\mathcal{P}_d(2t) = U_d(t)$, $\mathcal{C}_n(2t) = 2T_n(t)$ and the composition identity $U_{dn+n-1} = U_d(T_n)\,U_{n-1}$) to *Hall's conjecture* $\mathcal{C}_{n,d} = \mathcal{P}_d(\mathcal{C}_n)$, proved in [5]. Our route is different and elementary; we make no use of matching-polynomial combinatorics.

Corollary 2. $\displaystyle\sum_{d\geq 0} \mu_{d,C_n}(x)\, z^d = \frac{1}{1 - 2T_n(x/2)\, z + z^2},$ *and* $\displaystyle\sum_{r\geq 0} \Phi_{n,r}(x)\, z^r = \frac{(1-z)^2}{1 - 2T_n(x/2)\, z + z^2}.$

Corollary 3 (real-rootedness)**.** *All roots of $\Phi_{n,r}$ lie in $[-2, 2]$: with $x = 2\cos\theta$, χ_{C_n} vanishes at $\theta = 2\pi m/n$ and $\Psi_r = \sin(rn\theta)/\sin(n\theta)$ vanishes at $\theta = m\pi/(rn)$ with $r \nmid m$ (when $r \mid m$ the denominator vanishes too and $\Psi_r = \pm r \neq 0$); this leaves exactly $n(r-1)$ roots, matching* $\deg \Psi_r$.

Corollary 4 (Fibonacci–Lucas evaluation)**.** $\Phi_{n,r}(3) = (L_{2n} - 2)\, F_{2nr}/F_{2n}$*, since* $T_n(3/2) = L_{2n}/2$ *and* $U_{r-1}(L_{2n}/2) = F_{2nr}/F_{2n}$*.*

Corollary 5 (parity)**.** $\Psi_r(-x) = (-1)^{n(r-1)}\Psi_r(x)$*: the quotient is supported on exponents of a single parity, like a matching polynomial. (For bipartite* C_n *it is an even polynomial for every* r*; at* $r = 2$ *it is the classical matching polynomial* $2T_n(x/2)$ *of* C_n*, consistent with Godsil–Gutman [3].)*

2 Proof

Write $f(\ell) = 2T_\ell(Y) - 2$ with $Y = T_n(x/2)$; by the composition identity $T_{n\ell} = T_\ell \circ T_n$ this is the per-cycle factor in (1).

Step 1 (exponential formula). For any commutative ring R and $f : \mathbb{N} \to R$, the S_r-average $\Phi_r = \mathbb{E}_\sigma \prod_{\ell \in \mathrm{cycles}(\sigma)} f(\ell)$ satisfies

$$r\,\Phi_r = \sum_{k=1}^{r} f(k)\,\Phi_{r-k}, \qquad \Phi_0 = 1, \tag{2}$$

the coefficient form of $\sum_r \Phi_r z^r = \exp\bigl(\sum_\ell f(\ell) z^\ell/\ell\bigr)$. (Proof: group σ by cycle type λ with Cauchy's count $r!/z_\lambda$, mark a part, and re-index; see the Lean file `Paper2ExpFormula.lean` for the fully detailed version.)

Step 2 (Chebyshev generating function). With $f(\ell) = 2T_\ell(Y) - 2$,

$$\sum_{\ell \ge 1} f(\ell)\frac{z^\ell}{\ell} = -\log\bigl(1 - 2Yz + z^2\bigr) + 2\log(1-z) \quad \Longrightarrow \quad \sum_{r \ge 0} \Phi_r z^r = \frac{(1-z)^2}{1 - 2Yz + z^2}.$$

Extracting coefficients against $\sum U_r(Y) z^r = (1 - 2Yz + z^2)^{-1}$ gives $\Phi_r = U_r - 2U_{r-1} + U_{r-2}$, and the three-term recurrence $U_r + U_{r-2} = 2YU_{r-1}$ collapses this to $(2Y - 2)\,U_{r-1}(Y)$. □

In the formalization the analytic Step 2 is replaced by an equivalent finite argument: the closed form is shown to satisfy (2) directly (a second-difference induction), and (2) with $\Phi_0 = 1$ determines Φ over a characteristic-zero domain.

3 Graphs of first Betti number one

The proof above uses cycles in exactly one place: gauge-fixing a spanning path leaves a *single* permutation, because C_n has $|E| - |V| + 1 = 1$. The argument therefore applies verbatim to any connected G with $b_1(G) = |E| - |V| + 1 = 1$, a cycle with trees attached, and we record what it gives.

By [4] the expected characteristic polynomial of a random r-lift of *any* graph factors as $\chi_G \cdot \mu_{r-1,G}$; the content below is therefore not that factorization but a closed form for the factor $\mu_{d,G}$ itself. Question 6.5 of [4] asks which parts of the theory of the matching polynomial $\mu_{1,G}$ generalize to $\mu_{d,G}$; the following answers it for $b_1(G) = 1$.

Let C be the unique cycle of G and let $G - V(C)$ be the forest obtained by deleting its vertices. Write μ_H for the ordinary matching polynomial of a graph H, with $\mu_\emptyset = 1$. Define

$$V_0 = 1, \qquad V_1 = \mu_G, \qquad V_d = \mu_G\, V_{d-1} - \mu_{G-V(C)}^2\, V_{d-2}. \tag{3}$$

Theorem 6. *For every connected G with $b_1(G) = 1$ and every $d \geq 0$, $\mu_{d,G} = V_d$.*

For $G = C_n$ the deleted forest is empty, so $\mu_{G-V(C)} = 1$ and (3) is the recurrence for $U_d(\mu_{C_n}/2) = U_d(T_n(x/2))$: Theorem 1 is the case $\mu_{G-V(C)} = 1$. In general the attached trees enter only through $\mu_{G-V(C)}$; for the triangle with one pendant edge, $\mu_G = x^4 - 4x^2 + 1$ and $\mu_{G-V(C)} = x$.

The proof goes through the twisted characteristic polynomial. Let $e = uv$ be the unique non-tree edge, let $A_G(z)$ be the adjacency matrix of G with the (u, v) entry replaced by z and the (v, u) entry by z^{-1}, and set $F(x, z) = \det(xI - A_G(z))$.

Lemma 7. $F(x, z) = \mu_G(x) - \mu_{G-V(C)}(x)\,(z + z^{-1})$.

Proof. By the Sachs coefficient theorem $\det(xI - A)$ expands over elementary subgraphs, that is disjoint unions of edges and cycles. Since a permutation uses each of the two twisted entries at most once and $A_G(z)^\mathsf{T} = A_G(z^{-1})$, the determinant is of the form $P(x) + Q(x)(z + z^{-1})$. Replacing z by ± 1 and averaging kills exactly the elementary subgraphs whose cycle traverses e; as G has the single cycle C, those are the subgraphs containing C, so $P = \mu_G$, and the surviving terms contribute $Q = -\mu_{G-V(C)}$, a cycle component carrying the Sachs weight -2. □

Proof of Theorem 6. Gauge-fix a spanning tree: the r-lift is determined by one permutation $\sigma \in S_r$ on e, and its components are indexed by the cycles of σ, the component of an ℓ-cycle being the ℓ-fold cyclic cover G_ℓ. Decomposing the $\mathbb{Z}/\ell$ action into characters gives the classical factorization $\chi_{G_\ell}(x) = \prod_{j=0}^{\ell-1} F(x, \omega^j)$ with $\omega = e^{2\pi i/\ell}$ [24, 25]. With P, Q as in Lemma 7 and $y = -P/(2Q)$,

$$\chi_{G_\ell} = \prod_{j=0}^{\ell-1}\Big(P + 2Q\cos\tfrac{2\pi j}{\ell}\Big) = (-Q)^\ell\big(2T_\ell(y) - 2\big),$$

so by the exponential formula (2), with $w = -Qz$,

$$\sum_{r\geq 0}\Phi_{G,r}z^r = \exp\Big(\sum_{\ell\geq 1}\big(2T_\ell(y) - 2\big)\frac{w^\ell}{\ell}\Big) = \frac{(1-w)^2}{1 - 2yw + w^2},$$

and extracting coefficients as in Section 2 gives $\Phi_{G,r} = \chi_G \cdot (-Q)^{r-1}U_{r-1}(y)$. Comparing with $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$ [4] yields $\mu_{d,G} = (-Q)^d U_d(y)$, which satisfies (3). □

Since $U_d(y) = 2^d\prod_{k=1}^d(y - \cos\frac{k\pi}{d+1})$, the theorem can be restated as a product, $\mu_{d,G}(x) = \prod_{k=1}^d F\big(x, e^{ik\pi/(d+1)}\big)$, whence:

Corollary 8. *For $b_1(G) = 1$, $\mu_{d,G}$ is a product of d real-rooted polynomials, and its roots are exactly the eigenvalues of the Hermitian matrices $A_G\big(e^{ik\pi/(d+1)}\big)$, $1 \leq k \leq d$.*

This is the real-rootedness theorem of [4] for this family, with the roots identified. The localization itself — that a vanishing V_d forces $|\mu_G| \leq 2|\mu_{G-V(C)}|$, so that a root can never sit in a spectral gap — is machine-checked in Lean 4 as `cV_root_mem_band`, resting on `cU_ne_zero_of_one_le_abs`: the Chebyshev polynomial U_d has no root of absolute value at least one.

The roots can be located much more precisely than that. Since $b_1(G) = 1$ the group $\pi_1(G)$ is $\mathbb{Z}$, so the universal cover T of G *is* the $\mathbb{Z}$-cover, and T is a periodic decorated chain whose adjacency operator decomposes by Floquet–Bloch into the fibres $A_G(e^{i\theta})$. Hence

$$\operatorname{spec}(T) = \bigcup_\theta \operatorname{spec} A_G(e^{i\theta}) = \Big\{x :\ \exists\, s \in [-1,1],\ \mu_G(x) = 2s\,\mu_{G-V(C)}(x)\Big\},$$

a finite union of bands, together with a flat band at each common zero of μ_G and $\mu_{G-V(C)}$, where $F(x,\cdot)$ vanishes identically.

Corollary 9. *For $b_1(G) = 1$ the roots of $\mu_{d,G}$ are exactly the solutions of $\mu_G(x) = 2\cos\!\big(\frac{k\pi}{d+1}\big)\mu_{G-V(C)}(x)$ for $1 \le k \le d$. In particular every root of every $\mu_{d,G}$ lies in* $\operatorname{spec}(T)$, *and as $d \to \infty$ the roots equidistribute on* $\operatorname{spec}(T)$ *with respect to its density of states.*

Proof. Immediate from Theorem 6 in the form $\mu_{d,G} = \prod_k F(x, e^{ik\pi/(d+1)})$ and $F(x, e^{i\theta}) = \mu_G - 2\cos\theta\,\mu_{G-V(C)}$. The equidistribution is the arcsine law for $\cos\big(k\pi/(d+1)\big)$ transported through the fibre map. $\square$

Hall–Puder–Sawin place the roots of $\mu_{d,G}$ in the Ramanujan *interval* $[-\rho, \rho]$, ρ being the spectral radius of T. Corollary 9 places them in $\operatorname{spec}(T)$, which is a union of bands and is a proper subset of that interval whenever T has a spectral gap: no root of any $\mu_{d,G}$ can ever enter a gap. The two statements coincide exactly when the band structure is trivial, and that is precisely the case of a cycle, where $\mu_{G-V(C)} = 1$ and $\operatorname{spec}(T) = [-2, 2]$ is the whole interval. Attaching even a single pendant edge opens gaps, and they are not small (computed band structures, proportion of $[-\rho, \rho]$ that is a gap):

G	ρ	gap proportion
C_n	2	0%
triangle + pendant	2.170	36.5%
C_4 + pendant	2.136	45.9%
triangle + P_2	2.214	51.3%
$C_4 + P_3$	2.189	54.1%
triangle + $K_{1,3}$	2.514	65.9%

So the cycle is not merely the easiest case of Theorem 6: it is the degenerate one, the unique member of the family whose spectrum has no gaps and for which the Ramanujan interval is attained. Any cycle with a tree attached exhibits a band structure that no cycle can display.

The boundary is sharp. For $b_1(G) = b$ the lift is determined by b permutations, that is by an action of the free group F_b on $[r]$, whose orbits are the components of the lift. The exponential formula still applies, in the form $N_r = \sum_\ell \binom{r-1}{\ell-1} C_\ell N_{r-\ell}$ with $N_r = (r!)^b \Phi_{G,r}$ and C_ℓ the sum of χ over the transitive b-tuples on $[\ell]$; we have verified this for $b \le 3$. What fails is the atom. For $b = 1$ every transitive tuple gives the *same* cover G_ℓ, so $C_\ell = (\ell-1)!\,\chi_{G_\ell}$: one graph per ℓ, with χ_{G_ℓ} Chebyshev in ℓ. For $b \ge 2$ the transitive tuples number $(\ell-1)!$ times the count of index-ℓ subgroups of F_b and spread over many non-isomorphic covers with distinct spectra, so C_ℓ carries no such structure. Accordingly Theorem 6 is not merely unproved for $b \ge 2$: it is false there already at $d = 1$, as we checked for the theta graph, the bowtie, K_4, $K_{2,3}$ and two triangles joined by an edge.

Theorem 6 was verified by exact integer computation against the definition of $\mu_{d,G}$ as an average of matching polynomials over all d-covers, and against brute-forced lift averages, on thirteen named unicyclic graphs for $r \le 6$ and on a random sample of thirty-four for $r \le 5$; Lemma 7 was checked on eleven.

4 A conjecture for all graphs

Corollary 9 is proved by Floquet theory, which is available only because $\pi_1(G) = \mathbb{Z}$. Its *statement*, however, makes sense for every graph, and we know one case of it: at $b_1(G) = 1$ it is Corollary 9, for every d. We therefore propose:

Conjecture 10. *For every finite graph G and every $d \ge 1$, all roots of $\mu_{d,G}$ lie in* $\mathrm{spec}(T)$*, the spectrum of the universal cover of G.*

The biregular case of Conjecture 10 at $d = 1$ is not new. It is Problem 1 of Song, Fan and Miao [7], posed in 2023: they ask whether the matching polynomial of the incidence graph B_H of a (d,r)-regular hypergraph has no root μ with $0 < |\mu| < |\sqrt{d-1} - \sqrt{r-1}|$, and every (d,r)-biregular bipartite graph is such an incidence graph. Their Theorem 1.8 attaches a consequence: an affirmative answer would give every (d,r)-regular hypergraph a left-sided Ramanujan 2-covering. The only bound in that direction we know of is their Lemma 5.5, which goes the other way, $0 < \mu_\tau(B_H) \le \max\{\sqrt{d}, \sqrt{r}\}$. We are not aware of any proof, partial result or counterexample. Conjecture 10 extends their problem in two directions, to arbitrary graphs and to all d.

Question 6.3 of [4] asks the corresponding question for graph eigenvalues, and leaves it open: it distinguishes graphs whose nontrivial spectrum lies in the *interval* $[-\rho, \rho]$ from those whose nontrivial spectrum lies in $\mathrm{spec}(T)$, observes that the two notions separate exactly when T has a gap, and gives the biregular tree as the example. Conjecture 10 is the matching-polynomial form of that distinction.

The conjecture strengthens the root-localization theorem of [4], which places the roots in the interval $[-\rho, \rho]$ with ρ the spectral radius of T; the conjecture places them in $\mathrm{spec}(T)$ itself. The two agree whenever T has no spectral gap — in particular for regular G, where T is a regular tree and $\mathrm{spec}(T) = [-2\sqrt{q}, 2\sqrt{q}]$ is an interval. They differ exactly when T has a gap, and the smallest examples are biregular: for $K_{2,3}$ the universal cover is the $(3,2)$-biregular tree, with spectrum $\{0\} \cup \pm[\sqrt{2}-1, \sqrt{2}+1]$ and hence a gap $(0, \sqrt{2}-1)$ inside the Ramanujan interval. We have tested the conjecture on the biregular family, where the universal cover is an (a,b)-biregular tree with spectrum $\{0\} \cup \pm[\,|s-t|, s+t\,]$, $s = \sqrt{a-1}$, $t = \sqrt{b-1}$, so the gap is $(0, |s-t|)$. In each case $\mu_{d,G}$ was computed by averaging matching polynomials over all d-covers. No root ever lies in the gap; we also record the smallest nonzero $|\mathrm{root}|$ as a multiple of the gap edge, which measures how close the bound comes to being attained.

G	b_1	type	gap edge	d	min |root|/gap edge
$K_{2,3}$	2	$(3,2)$	0.4142	$1,2,3,4$	$2.72,\ 2.19,\ 1.94,\ 1.78$
$K_{2,4}$	3	$(4,2)$	0.7321	$1,2,3$	$1.93,\ 1.65,\ 1.52$
$K_{2,5}$	4	$(5,2)$	1.0000	$1,2,3$	$1.66,\ 1.46,\ 1.37$
$K_{3,4}$	6	$(4,3)$	0.3178	$1,2$	$3.04,\ 2.31$
$K_{3,5}$	8	$(5,3)$	0.5858	$1,2$	$2.10,\ 1.73$
subdivision of K_4	3	$(3,2)$	0.4142	$1,2$	$1.97,\ 1.67$

All of these have $b_1 \geq 2$, so none is covered by Corollary 9. It is worth recording that even $d = 1$ is not formal. There $\mu_{1,G}$ is the ordinary matching polynomial, whose roots are eigenvalues of the path tree of G (Godsil), and the path tree embeds in T; but eigenvalues of a finite subtree of T need not lie in spec(T) when T has a gap. For the $(3,2)$-biregular tree, whose gap is $(0, \sqrt{2}-1)$, the path P_8 is a subtree with eigenvalue $2\cos(4\pi/9) = 0.3473$ inside that gap. So the $d = 1$ case does not follow from the path-tree argument. It is nevertheless a theorem for an infinite family.

4.1 Subdivisions: the case of minimum degree two

Theorem 11. *Let H be D-regular, $D \geq 2$, and let $S(H)$ be its subdivision, so that the universal cover of $S(H)$ is the $(D,2)$-biregular tree with spectral gap $\big(0, \sqrt{D-1}-1\big)$. Then every nonzero root of $\mu_{S(H)}$ has absolute value at least $\sqrt{D-1}-1$; that is, Conjecture 10 holds for $S(H)$ at $d = 1$. The constant is the spectral gap of the universal cover; we do not claim it is attained, and by the strictness of Heilmann–Lieb on finite graphs it is not.*

Theorem 11 settles the case $\min(d,r) = 2$ of Problem 1 of [7]. Their uniformity condition allows $r \geq 2$, and they observe that when H is a simple graph the incidence graph B_H is exactly the subdivision $S(H)$; so their question at $r = 2$ asks precisely whether $\mu_{S(H)}$ has no root μ with $0 < |\mu| < |\sqrt{D-1}-1|$. By their Theorem 1.8 this gives every D-regular graph, viewed as a $(D,2)$-regular hypergraph, a left-sided Ramanujan 2-covering. The case $d = 2$ follows as well, since B_H is a bipartite graph and μ_{B_H} does not distinguish its two sides: a $(2,r)$-regular hypergraph has for incidence graph the subdivision of an r-regular graph, and the quantity $|\sqrt{d-1} - \sqrt{r-1}|$ is symmetric in d and r. The case $\min(d,r) \geq 3$ of their problem remains open, as does Conjecture 10 for $\min(a,b) \geq 3$.

Proof. By the subdivision identity $\mu_{S(H)}(x) = x^{|E|-|V|}\,\mu_H(x^2 - D)$ [6], the nonzero roots of $\mu_{S(H)}$ are the x with $x^2 = D + \theta$ for θ a root of μ_H. Heilmann–Lieb gives $\theta \geq -2\sqrt{D-1}$, so with $s = \sqrt{D-1}$ we get $x^2 \geq s^2 + 1 - 2s = (s-1)^2$, whence $|x| \geq s - 1$. Sharpness: for cubic H of large girth $\theta_{\min} \to -2\sqrt{2}$, so the bound is approached. □

The inequality step is machine-checked in Lean 4 as `subdivision_gap`. The theorem is exactly Heilmann–Lieb for H transported through the subdivision identity, and the numbers match: for $H = K_4$, $\mu_{K_4} = x^4 - 6x^2 + 3$ gives smallest positive root $\sqrt{3 - \sqrt{3+\sqrt{6}}} = 0.8158$, against the gap edge 0.4142; the subdivisions of $K_{3,3}$, the prism, Q_3, the Wagner, Petersen, Heawood and Möbius–Kantor graphs give $0.701, 0.714, 0.650, 0.647, 0.607, 0.563, 0.551$, decreasing toward the gap edge with the girth and never reaching it.

For $K_{2,3}$ the computation was pushed to $d = 6$ using $\mu_{d,G} = \mathbb{E}[\chi_H]/\chi_G$ over $(d+1)$-covers, which is polynomial time per cover, together with the reduction of the first free permutation to one representative per cycle type; the ratios continue 1.68 at $d = 5$ and no root in the gap at $d = 6$. That last case is a caution worth recording: in floating point it appeared to produce roots at ± 0.055, well inside the gap, which an exact integer recomputation showed to be an artefact of evaluating a degree-30 polynomial built from 35×35 characteristic polynomials in double precision.

The last column decreases monotonically in d in every case, which is what the conjecture predicts: by Corollary 9 in the $b_1 = 1$ case the roots equidistribute on $\mathrm{spec}(T)$, so they should approach the gap edge without crossing it. The evidence is therefore not merely that the bound holds but that it is approached, though no computation here reaches the regime where the ratio is close to 1.

4.2 The biregular case is one interval, and we have one end of it

Let G be (a, b)-biregular bipartite, $s = \sqrt{a-1}$, $t = \sqrt{b-1}$. Since G is bipartite, $\mu_G(x) = x^{\varepsilon} f(x^2)$ for a polynomial f and $\varepsilon \in \{0, 1, 2, \dots\}$. Put

$$c = (a-1) + (b-1), \qquad m = (a-1)(b-1), \qquad g(z) = f(z + c).$$

The universal cover has spectrum $\{0\} \cup \pm[\,|s-t|, s+t\,]$, and $(s \pm t)^2 = c \pm 2\sqrt{m}$, so

$$x \in \mathrm{spec}(T) \ \setminus \{0\} \iff x^2 - c \in \left[-2\sqrt{m},\ 2\sqrt{m}\right].$$

The change of variable therefore turns Conjecture 10, for biregular G, into a single clean statement:

$$\text{every root of } g \text{ lies in } \left[-2\sqrt{m},\ 2\sqrt{m}\right], \tag{4}$$

which is a Heilmann–Lieb interval of effective degree $m + 1$.

It is worth recording what this says about the matching generating polynomial $M_G(w) = \sum_k m_k w^k$, since that, rather than μ_G, is the object of [10]. From $f_G(y) = y^p M_G(-1/y)$ the roots correspond by $w = -1/y$, so a lower bound on the least positive root of f_G is an upper bound on the modulus of the roots of M_G. Writing $Q_G(z) = M_G(-z) = \sum_k (-1)^k m_k z^k$, the reciprocal-and-reflect of f_G:

$$\text{Conjecture 10 at } d = 1 \iff \text{every root of } Q_G \text{ lies in } \Big(0,\ (s-t)^{-2}\Big]. \tag{5}$$

Heilmann–Lieb, in the same variable, says every root of Q_G is at least $\left(2\sqrt{\Delta - 1}\right)^{-2}$ in modulus. The two statements bound the roots of one polynomial from the two sides, so (4) and (5) are the same dichotomy seen in reciprocal coordinates. This is a restatement and nothing more, but it is the coordinate in which the classical machinery is written, and Q_G is the polynomial whose deletion recursion $Q_G = Q_{G-v} - z \sum_{u \sim v} Q_{G-v-u}$ is formalized in the accompanying Lean development. The upper end of (4) is known: the path-tree argument of Godsil bounds the roots of μ_G by the spectral radius of the path tree, which for an (a, b)-biregular graph with $a, b \geq 2$ is at most $s + t$ [19, 20]. It is worth noting that [10] proved the corresponding statement only in the regular case; the irregular case is due to Godsil and to [26]. The lower end is the conjecture. *They are the two ends of one interval*, so what is missing is exactly half of a two-sided Heilmann–Lieb theorem for biregular bipartite graphs.

5 The coefficient ladder

This also explains the boundary between what is proved and what is not, and the explanation is exact.

Theorem 12. *Let G be (a, b)-biregular bipartite with parts of sizes $p \le q$, degree a on the p-side and b on the q-side, and let $g(z) = f_G(z + c)$ with $c = (a - 1) + (b - 1)$. Then*

$$[z^{p-1}]\, g \;=\; p\,(b-2).$$

Consequently the roots of g have mean $-(b-2)$, and g is the matching polynomial of a graph if and only if $b = 2$.

Proof. Counting edges by their smaller side gives $pa = qb$, so $p \le q$ forces $a \ge b$ and $b = \min(a, b)$. The polynomial f_G is monic of degree p, and its next coefficient is $-m_1$, where m_1 is the number of 1-matchings, that is $|E| = pa$. Hence

$$[z^{p-1}]\, f_G(z + c) = \binom{p}{1} c - pa = p(a + b - 2) - pa = p(b - 2).$$

The mean of the roots is $-[z^{p-1}]g/p = -(b-2)$. A matching polynomial of degree p involves only every other power of z, so its coefficient at z^{p-1} vanishes; thus $b = 2$ is necessary. It is sufficient: for $b = 2$ the graph is $S(H)$ for an a-regular H and $g = \mu_H$ by [8]. $\square$

So for $b = 2$ the polynomial g *is* a matching polynomial, Heilmann–Lieb applies verbatim, and one recovers Theorem 11 and the subdivision identity as the case $c = (a - 1) + 1$ of the change of variable. For $b \ge 3$ it provably is not, and Theorem 12 says more than that it fails: it locates the failure. The roots of g are not centred at the band centre but sit a distance $b - 2$ below it, towards the lower band edge — which is exactly the end of (4) that is open. The asymmetry of the problem and the departure from the matching-polynomial class are the same phenomenon, measured by the single quantity $b - 2$. Examples:

$$g_{K_{3,4}} = z^3 + 3z^2 - 9z - 19, \quad g_{K_{3,5}} = z^3 + 3z^2 - 12z - 24, \quad g_{K_{4,5}} = z^4 + 8z^3 - 6z^2 - 128z - 139,$$

with subleading coefficients $3 = 3(3 - 2)$, $3 = 3(3 - 2)$ and $8 = 4(4 - 2)$ as the theorem requires.

The same method determines more than the one coefficient, and shows that the graph itself does not enter until surprisingly late.

Theorem 13. *With notation as in Theorem 12, the coefficients of z^p, z^{p-1}, z^{p-2} and z^{p-3} in g depend only on (p, a, b), not on G. Explicitly the first three are 1, $p(b-2)$ and $\frac{p}{2}\big[(p-1)(b-2)^2 - a(b-1)\big]$.*

Proof. The coefficients of g up to z^{p-j} are determined by $m_1, \dots, m_j$, so it suffices that m_1, m_2, m_3 are functions of (p, a, b) alone. Write $N = |E| = pa = qb$. Then $m_1 = N$. Two distinct edges of a bipartite graph meet in at most one vertex, so $m_2 = \binom{N}{2} - p\binom{a}{2} - q\binom{b}{2}$, which after substituting $q = pa/b$ gives $m_2 = \frac{pa}{2}(pa - a - b + 1)$; the stated value of $[z^{p-2}]g$ follows from $\binom{p}{2}c^2 - (p-1)Nc + m_2$ with $c = a + b - 2$.

For m_3, let L be the graph on E in which two edges are adjacent when they meet, so m_3 is the number of independent 3-sets of L. Every vertex of L has degree $(a - 1) + (b - 1) = c$.

Three pairwise meeting edges of a bipartite graph must share a vertex, since otherwise they form a triangle in G; hence the number of triangles of L is $t = p\binom{a}{3} + q\binom{b}{3}$, and the number of 2-edge paths is $N\binom{c}{2} - 3t$. The standard count $\binom{N}{3} - \frac{Nc}{2}(N-2) + \big(N\binom{c}{2} - 3t\big) + 2t$ therefore depends only on (p, a, b). $\square$

The graph enters at the very next coefficient, and it does so through a single invariant, with coefficient one.

Theorem 14. *Let $C_4(G)$ denote the number of 4-cycles of G. Then*

$$[z^{p-4}]\, g \;=\; U(p, a, b) + C_4(G),$$

where, writing $\beta = b - 2$ and $A = a(b-1)$,

$$U = \binom{p}{4}\beta^4 - \tfrac{3}{2}A\beta^2\binom{p}{3} + \frac{A\,(3A + 4\beta^2)}{12}\binom{p}{2} - \frac{A\,(9A + 2b^2 - 14b + 14)}{24}\,p.$$

Proof. As in Theorem 13, m_4 is the number of independent 4-sets of L, and inclusion–exclusion expresses it through the *non-induced* counts in L of the ten graphs on at most four vertices with no isolated vertex. Nine of the ten are universal, by the clique structure of L: each vertex of L is a cell of the $p \times q$ biadjacency matrix, and two cells are adjacent exactly when they share a row or share a column, never both, so every edge of L has a well-defined type.

Only the count of C_4 in L is not universal. Reading the cyclic word of types around a four-cycle of L, the words $RRRC$ and $RCCC$ are impossible, since three consecutive edges of one type force all four cells into a single line and then the fourth edge has that type too; and $RRCC$ is impossible, since it would require a cell sharing a column with two distinct cells of one row. This leaves $RRRR$ and $CCCC$, whose four cells lie in one line and which therefore number $3\big[p\binom{a}{4} + q\binom{b}{4}\big]$, and $RCRC$, whose four cells are exactly the four entries of a 2×2 all-ones submatrix, that is a 4-cycle of G, each contributing exactly one. Substituting and applying the shift $[z^{p-4}](z + c)^{p-k} = \binom{p-k}{p-4}c^{4-k}$ gives the stated U. $\square$

Together with Theorem 13 this determines the top five coefficients of g completely:

$$[z^{p-1}]g = p\beta, \qquad [z^{p-2}]g = \binom{p}{2}\beta^2 - \tfrac{1}{2}Ap,$$

$$[z^{p-3}]g = \binom{p}{3}\beta^3 - A\beta\binom{p}{2} + \tfrac{1}{6}A\beta p, \quad [z^{p-4}]g = U + C_4(G).$$

For the Heawood graph ($p = 7$, $a = b = 3$, $C_4 = 0$) the last reads $35 - 315 + 231 - 77 = -126$. So the conjecture is insensitive to the graph until girth enters, and it enters at the first opportunity: four pairwise disjoint edges are first constrained by a 4-cycle.

Theorem 13 has a consequence for what a proof of Conjecture 10 can look like. The universal coefficients determine the first two power sums of the roots of g, namely $\sum \rho_i = -p(b-2)$ and $\sum \rho_i^2 = p\big[(b-2)^2 + a(b-1)\big]$, and those two numbers already imply a bound on the least root.

Proposition 15. *Every root of g satisfies $\rho \geq -(b-2) - \sqrt{(p-1)a(b-1)}$, and this is the sharpest bound implied by the coefficients of z^p, z^{p-1}, z^{p-2}.*

Proof. The roots are real. Fix a root ρ and apply Cauchy–Schwarz to the other $p-1$: $\big(\sum \rho_i - \rho\big)^2 \le (p-1)\big(\sum \rho_i^2 - \rho^2\big)$. With $e = \sum \rho_i$ and $S = \sum \rho_i^2$ this is $p\rho^2 - 2e\rho + e^2 - (p-1)S \le 0$, so $\rho \ge \big[e - \sqrt{(p-1)(pS-e^2)}\big]/p$. Substituting the two power sums gives $pS - e^2 = p^2 a(b-1)$ and the stated bound. Sharpness is the equality case of Cauchy–Schwarz. □

The bound of Proposition 15 grows like $\sqrt{p}$, whereas the conjectured bound $-2\sqrt{m}$ does not depend on p at all. So for all but the smallest graphs the moment bound is far weaker, and since by Theorem 13 the coefficients it uses do not see G, no argument resting on them can distinguish one (a,b)-biregular graph from another. Any proof of Conjecture 10 must therefore use m_4 or beyond, that is, it must be sensitive to the cycle structure of G. This is consistent with the place the conjecture occupies: bounds of the shape $2\sqrt{(a-1)(b-1)}$ are non-backtracking bounds, and the non-backtracking operator of a bipartite graph is exactly the object that records cycles. That object can be exhibited, but only in averaged form, and the reason for the averaging is not incidental.

6 An expected non-backtracking polynomial

Lemma 16 (Chernyak–Chertkov; Watanabe–Fukumizu)**.** *Let G be any graph, A_s its adjacency matrix with edges signed by $s \in \{\pm 1\}^E$, and D the degree matrix. Then, with s uniform,*

$$\mathbb{E}_s \det\big(I + u^2(D-I) - uA_s\big) = \sum_M (-u^2)^{|M|} \prod_{v \notin V(M)} \big(1 + (d_v - 1)u^2\big),$$

the sum being over all matchings M of G. If G is (a,b)-biregular bipartite with $P = a-1$ and $Q = b-1$, the right side is $\mu_G(\lambda)$ up to normalization at $\lambda = \sqrt{(1+Pu^2)(1+Qu^2)}/u$, and then, with $t = u^2$,

$$\lambda^2 - c = \frac{1}{t} + mt. \tag{6}$$

The identity is not new and we claim none of it. Chernyak and Chertkov [14, Eqs. (2), (6)] prove that for an arbitrary matrix H the average of $\det H(\sigma)$ over $\mathbb{Z}_2$ edge gaugings equals a monomer–dimer sum with weights $w_a = H_{aa}$ and $w_{ab} = -H_{ab}H_{ba}$; taking $H = I + u^2(D-I) - uA$ gives the display, since then $w_a = 1 + (d_a - 1)u^2$ and $w_{ab} = -u^2$. Equivalently, the averaging step alone is the Godsil–Gutman expansion [3], valid for any diagonal matrix in place of $I + u^2(D-I)$, and the right-hand side with those degree-dependent weights is the monomer–dimer partition function $\Xi_G(-u^2, \lambda)$, $\lambda_v = 1 + (d_v - 1)u^2$, of Watanabe and Fukumizu [18, Thm. 5.5]; on a $(q+1)$-regular graph they reduce it to $u^{|V|}\mu_G(1/u + qu)$ [18, Eq. (5.3)], attributing the regular case to Nagle — an attribution we take from [18], not having consulted the original — and record the consequent band statement, that the roots then lie on the circle of radius $1/q$. Only (6), the biregular substitution, is not in those sources.

What we do wish to point out is how the left-hand side should be read. By the Ihara–Bass formula for signed graphs, $\mathbb{E}_s \det(I - uW_s)$ is $(1-u^2)^{|E|-|V|}$ times the left side above, so the polynomial in Lemma 16 is an *expected non-backtracking characteristic polynomial.* This reading is absent from the sources just cited, and not by oversight in an obvious direction: [14] obtain the average but explicitly set the Ihara spectral parameter to zero, and [18] display the Bass matrix $I - uA_G + u^2(D_G - I)$ next to their Theorem 5.9 — a sum over deletions of vertex-disjoint cycles — and remark in print that its relation to the graph zeta function is unknown. The sign average is exactly what annihilates those cycle terms. The closest antecedent we have found is Watanabe and Chertkov [15, Thm. 4], who do average a non-backtracking determinant over ± 1 edge signs and convert it by Ihara–Bass, but for the permanent, that is for perfect matchings on $K_{N,N}$, with no degree-dependent vertex weights.

The band condition has a clean reading on that polynomial, and it sharpens a published bound. By (6), a root β of ω_G coming from a nonzero root λ of μ_G satisfies $m\beta^2 - \rho\beta + 1 = 0$ with $\rho = \lambda^2 - c$, so the two roots of that quadratic have product $1/m$; they are complex conjugates, hence both of modulus $m^{-1/2}$, exactly when $|\rho| \leq 2\sqrt{m}$, and otherwise real with one inside and one outside that circle. Watanabe and Fukumizu prove [18, Cor. 5.8] that every root of ω_G lies in the annulus $1/(d_M - 1) \leq |\beta| \leq 1/(d_m - 1)$, which for an (a, b)-biregular graph reads $1/(a-1) \leq |\beta| \leq 1/(b-1)$. Conjecture 10 at $d = 1$ asserts that all roots except those coming from $\lambda = 0$ lie on the single circle $|\beta| = m^{-1/2}$, the geometric mean of the two radii they bound by. The roots at $\lambda = 0$ are $\beta = -1/(a-1)$ and $\beta = -1/(b-1)$, so they sit at the two ends of that annulus, which is why it is attained and cannot be narrowed without excluding them.

Since $1/t + mt = 2\sqrt{m}\cos\varphi$ exactly when $|t| = 1/\sqrt{m}$:

Corollary 17. *For (a, b)-biregular bipartite G, Conjecture 10 at $d = 1$ holds for G if and only if every root u of the expected non-backtracking characteristic polynomial of G, other than the trivial ones, satisfies $|u| = m^{-1/4}$.*

So the conjecture is a Ramanujan condition after all, but for an *expected* polynomial rather than for a spectrum. That is forced. If μ_G were, up to the substitution and prefactor above, the determinant of a single locally described operator on G, its coefficients could be recovered by interpolation from polynomially many determinant evaluations; but those coefficients are the matching numbers m_k, whose computation is $\#P$-hard even for planar graphs of bounded degree [17]. No unaveraged Bass-type identity for the matching polynomial can therefore exist unless FP = #P. This is also why the roots of μ_G are not the spectrum of any operator on G, and hence why the available technology for such statements is the method of interlacing families [26], which works with expected polynomials, rather than a spectral argument.

7 How close the bound comes to being attained

Proposition 15 also explains why the interval is not filled. The mean square of the roots is $\big[(b-2)^2 + a(b-1)\big]$, so the fraction of the band $[-2\sqrt{m}, 2\sqrt{m}]$ occupied in mean square is

$$\frac{(b-2)^2 + a(b-1)}{4(a-1)(b-1)},$$

which is 7/16 for $(a, b) = (3, 3)$, 1/2 for $(2, 2)$, 3/8 for $(4, 3)$, and tends to 1/4 as $a \to \infty$. The bulk of the spectrum sits well inside the band; only the extremes are at issue, which is why moments cannot decide the question.

The extremes behave as Theorem 12 predicts. Writing the least and greatest roots as percentages of the two band edges:

G	girth	least root	greatest root
$K_{3,3}$	4	89.6%	57.2%
$K_{4,4}$	4	94.6%	56.6%
circulant $6+6$	4	96.4%	74.2%
Heawood	6	97.5%	79.9%
Pappus	6	98.6%	83.4%
C_{12}	6	96.6%	96.6%
$3C_4$	4	70.7%	70.7%

The first block has $b = 3$, and the least root is consistently far closer to its edge than the greatest is to its — the roots are displaced downwards by $b - 2$, exactly as Theorem 12 requires. The second block has $b = 2$, the displacement vanishes, and the two ends are used equally. That is what the table shows, and it is all it shows.

It does not bear on sharpness, for a reason worth stating because it is easy to miss. Every graph in the table has $a = b$, and there $c = 2(a-1) = 2\sqrt{m}$, so $\rho = y - c \geq -c = -2\sqrt{m}$ holds automatically from $y = x^2 \geq 0$: the lower bound is an identity, not a theorem. This is the triviality that Song, Fan and Miao note for $d = r$ [7]. The percentages above therefore measure only how close the least root of μ_G comes to 0, and 100% would require $\mu_G(0) = 0$, that is a graph with no perfect matching, which by Kőnig's theorem no regular bipartite graph is. The conjecture has content exactly when $a \neq b$, where $c > 2\sqrt{m}$ strictly by the arithmetic–geometric mean inequality.

Where it has content the bound holds with room to spare, and is not approached. For $(4,2)$-biregular graphs the conjecture asserts $y \geq c - 2\sqrt{m} = 4 - 2\sqrt{3} = 0.5359$; a girth-4 family plateaus at $y_{\min} = 1.1716$ and a girth-8 family at 0.7392. In the trivial regime the same plateauing is visible, and there the limit is an algebraic number one can name. For a cubic bipartite circulant on $n+n$ vertices with connection set S, the matching counts satisfy $f_n(y) = \operatorname{tr} A(y)^n$ for a transfer matrix $A(y)$ with entries linear in y, so by the theorem of Beraha, Kahane and Weiss the roots accumulate where the two dominant eigenvalues of $A(y)$ have equal modulus. That happens by a genuine collision of two real eigenvalues — below y^* the pair is real and distinct, above it a complex-conjugate pair — so the locus is cut out by $\operatorname{disc}_\lambda \chi_{A(y)} = 0$ and y^* is the left edge of the accumulation band. For $S = \{0, 1, 2\}$, girth 4,

$$\operatorname{disc}_\lambda \chi = 4y^2\big(y^4 - 12y^3 + 46y^2 - 84y + 5\big),$$

and y^* is the least positive root of that quartic, which is irreducible over $\mathbb{Q}$; then $y^* = 0.0615663466041589\ldots$ and the limit is $100 - 25y^* = 98.4608413349\ldots\%$ of the lower edge. For $S = \{0, 1, 3\}$, girth 6, the discriminant is $y^3F(y)$ with F irreducible of degree 17, giving $y^* = 0.0059255959422562\ldots$ and $99.8518601014\ldots\%$. Both stop strictly short of 100%. The finite-n values approach these from above with a $1/n^2$ band-edge law, and $n^2\big(y_{\min}(n) - y^*\big)$ is constant to five digits at $n = 200, 300, 500$, which identifies the extrapolated plateau with the algebraic number.

One half of the limit is certified exactly, without appeal to Beraha–Kahane–Weiss. On the whole of $[0, y^*)$ the largest eigenvalue modulus of $A(y)$ is attained by exactly one eigenvalue, which is real, simple and negative: the discriminant has no zero there, so real branches stay real; no two eigenvalues are negatives of one another, by a resultant of the even and odd parts of χ which has no zero on $(0, y^*]$; and for the octic a separating radius 5/4, which no eigenvalue ever attains, together with one exact Schur count, pins the number of eigenvalues outside it at two. Hence $|f_n| \geq |\lambda_1|^n(1-(N-1)(1+\delta)^{-n})$ on any compact subinterval, giving $\liminf_n y_{\min}(n) \geq y^*$. The reverse inequality does use Beraha–Kahane–Weiss: just above y^* the two branches form a conjugate pair of equal and maximal modulus, and one needs that theorem to conclude that the zeros accumulate there. This is what one should expect: a family with a fixed local structure has a fixed Benjamini–Schramm limit, which is not a tree, so its matching measure retains a gap at 0. Non-attainment of this kind was already visible to Heilmann and Lieb, who tabulate the largest zero against their own bound for six lattices — 11.24 against 12, 17.86 against 20, 41.32 against 44 — and remark that equality "cannot be expected to be true in general" [10, Table 1 and Remark 4.3]. Girth does help monotonically — at 30 vertices, girth 4, 6 and 8 give 98.166%, 99.311% and 99.410% — but no family of bounded girth can approach the edge, and every cubic bipartite circulant has girth at most 6, since the closed hexagon $0, s_1, s_1-s_2, s_1-s_2+s_3, s_3-s_2, s_3$ is always present. We therefore make no claim that $2\sqrt{m}$ is attained or approached. The roots of these circulants do satisfy (4), with maxima $4.06, 4.48, 6.26$ against bounds $4.90, 5.66, 6.93$.

8 The multivariate polynomial and the cavity recursion

The polynomials g admit a clean description. Since a k-matching leaves $p-k$ vertices of A unmatched,

$$f_G(y) = \sum_k (-1)^k m_k\, y^{p-k} = \sum_M (-1)^{|M|} \prod_{v \in A \setminus V(M)} y = \mathcal{M}\big(G;\, x_A = y,\ x_B = 1\big), \tag{7}$$

the multivariate matching polynomial of Heilmann and Lieb under the asymmetric specialization that puts the variable on one side of the bipartition and 1 on the other. So the g are not exotic: they are shifts of matching polynomials in the multivariate sense. This is a reformulation and not an advance: for bipartite G every edge of a matching covers one vertex of each side, so $\mathcal{M}(G; x_A = y, x_B = z)$ depends only on the product yz, and the asymmetric specialization therefore carries exactly the information of the univariate matching polynomial. In particular no quantitative Heilmann–Lieb theorem for non-uniform specializations can help here, and none is available: the multivariate bounds of [10] apply a single graph-wide threshold to every variable.

Deleting a vertex now gives two different recursions, because only A-vertices carry the variable:

$$v \in A: \quad f_G = y\, f_{G-v} - \sum_{u \sim v} f_{G-v-u}, \qquad u \in B: \quad f_G = f_{G-u} - \sum_{v \sim u} f_{G-u-v},$$

whence the cavity recursion $R_A = y - \sum 1/R_B$, $R_B = 1 - \sum 1/R_A$. On the (a,b)-biregular tree, with $P = a-1$, $Q = b-1$, its fixed point satisfies $R^2 + (P-Q-y)R + yQ = 0$, of discriminant

$$D(y) = (P-Q-y)^2 - 4yQ, \qquad D(y) = 0 \iff y = (\sqrt{P} \pm \sqrt{Q})^2.$$

The two roots are exactly the spectrum edges in the variable $y = x^2$, and $D < 0$ strictly between them: the fixed point is complex precisely on the spectrum and real precisely off it, as it must be. Both recursions and the discriminant have been checked exactly.

Conjecture 10 for biregular G is therefore the statement that the asymmetric recursion, started from the leaves of the path tree, never produces $R = 0$ while $D > 0$. The natural invariant region is

$$R_A \le -\alpha < 0, \qquad R_B \in [\,1,\ 1 + Q/\alpha\,],$$

for a parameter $\alpha > 0$. A B-vertex with children in the region has $R_B = 1 + \sum 1/|R_A| \in (1, 1 + Q/\alpha]$, and a B-leaf carries $R_B = 1$, exactly the lower boundary. An A-vertex with its full complement of P children has $R_A \le y - P\alpha/(\alpha + Q)$, so the region is closed precisely when $y \le h(\alpha)$, where $h(\alpha) = \alpha(P - Q - \alpha)/(\alpha + Q)$. The following is elementary and is machine-checked in Lean 4 (`region_closure_bound`, `gap_edge_identity`):

$$(s-t)^2(\alpha + t^2) - \alpha(s^2 - t^2 - \alpha) = \big(\alpha - t(s-t)\big)^2, \tag{8}$$

so $h(\alpha) \le (s-t)^2$ for every α, with equality at $\alpha = t(s-t) = \sqrt{PQ} - Q$. The region therefore closes for every y strictly below the gap edge and for no y above it: the invariant region is calibrated exactly to the spectrum, and (8) is the reason.

9 The reflection, and why the band edges are never roots

One structural fact explains why $b = 2$ is the case that closes, and it is a reflection. The subdivision identity $\mu_{S(H)}(x) = x^{|E|-|V|}\,\mathcal{LM}(H, x^2)$ is due to Yan and Yeh [8], who also treat the regular and semiregular cases; see [6], Corollary 2.3, for a restatement through the multivariate matching polynomial. For H D-regular it reads $f_G(y) = \mu_H(y - D)$, and $\mu_H(-z) = (-1)^{|V(H)|}\mu_H(z)$, so

$$f_G(2c - y) = (-1)^p f_G(y), \qquad c = (a-1) + (b-1), \tag{9}$$

and c is the midpoint of $[(s-t)^2, (s+t)^2]$ because $(s \pm t)^2 = c \pm 2\sqrt{m}$. The roots of f_G are therefore symmetric about the centre of the band, so for $b = 2$ the lower bound in (4) is *equivalent* to the upper one, and Theorem 11 is that equivalence. The reflection (9) fails as soon as both degrees exceed 2: it fails for $K_{3,4}$, $K_{3,5}$ and already for the regular $K_{3,3}$, so regularity does not rescue it. Since $b = 2$ holds exactly when G is the subdivision of an a-regular multigraph, (9) describes the whole of the phenomenon, and for $b \ge 3$ the lower bound is not a repackaging of the upper bound but carries information the upper bound does not. We note that (9) has no adjacency analogue: the companion identity $\varphi(S(H), x) = x^{|E|-|V|}\varphi(H, x^2 - D)$ holds as well, but $\varphi(H, \cdot)$ has a parity only for bipartite H, whereas μ_H always does. For the Petersen graph the matching-side values are symmetric about $D = 3$ and the adjacency-side values are not.

A consequence of the first is that the band edges are never roots, by an argument of a different kind.

Proposition 18. *Let G be (a, b)-biregular bipartite with $a, b \ge 2$, and suppose $m = (a-1)(b-1)$ is not a perfect square. Then neither $(s-t)^2$ nor $(s+t)^2$ is a root of f_G.*

Proof. f_G has integer coefficients and $(s \mp t)^2 = (a + b - 2) \mp 2\sqrt{m}$, so for m not a square the two band edges are conjugate over $\mathbb{Q}$; if one is a root of f_G, so is the other. Heilmann and Lieb already give a *strict* bound, $|a| < 2\sqrt{B_1}$ with $B_1 = \Delta - 1$ for unit edge weights [10, Thm. 4.3], but at the max-degree radius, which for $a \neq b$ exceeds $s+t$; strictness at the biregular radius therefore has to be argued separately. Every root of μ_G is bounded by the spectral radius of a path tree $T_u(G)$ [19], and $T_u(G)$ is finite and embeds in the universal cover of G, the (a,b)-biregular tree $T_{a,b}$, with degrees bounded by those of $T_{a,b}$. Now $\rho(T_{a,b}) = s+t$ is the supremum of ρ over finite subtrees of $T_{a,b}$, and that supremum is not attained: any finite subtree is a proper subgraph of a larger connected finite subtree, and ρ is strictly monotone under proper subgraph containment for finite connected graphs. Hence $\rho(T_u(G)) < s+t$, so $(s+t)^2$ is not a root of f_G, and therefore neither is $(s-t)^2$. □

The hypothesis on m is needed for the argument, not merely for convenience: when m is a square the two band edges are rational and independent, and conjugacy says nothing. The smallest such case is $(a,b) = (5,2)$, where $m = 4$ and the edges are 1 and 9. There the conclusion nevertheless holds, but for a different reason — the Heilmann–Lieb bound is strict for finite graphs, so Theorem 11 already excludes the lower edge. Proposition 18 is the statement for the cases those methods do not reach.

Evaluating f_G in $\mathbb{Z}[\sqrt{m}]$ confirms this and exhibits the conjugacy: for $K_{3,4}$, $K_{3,5}$, $K_{3,6}$, $K_{4,5}$ the two edge values are $53 \mp 30\sqrt{6}$, $72 \mp 40\sqrt{8}$, $91 \mp 50\sqrt{10}$, $1877 \mp 512\sqrt{12}$. For $b = 2$ the irrational parts vanish, as (9) requires. Proposition 18 is worth isolating because it uses no induction on vertex deletion, and so is not subject to the obstruction of Section 13; it is the only tool we have found with that property, although it controls only the edges and not the interior of the gap.

10 The operator form

10.1 The mixed characteristic polynomial

The multivariate object that does carry the biregular information is the mixed characteristic polynomial of Marcus, Spielman and Srivastava: with $P_k = \operatorname{diag}(\mathbf{1}_{N(k)})$ of rank b and $\sum_k P_k = aI_p$, one has $\nu_G = \mu[P_1, \ldots, P_q]$, since $\sum_k z_k P_k$ is diagonal and each ∂_{z_k} must strike a distinct linear factor, so the surviving terms are the injections, that is the matchings. This is the stability route done properly, and it locates the shortfall quantitatively: a barrier argument that sees only the rank, the trace and $\sum_k P_k = aI$ is reading the Marchenko–Pastur law, and yields the interval $\left[(\sqrt{a}-\sqrt{b})^2, (\sqrt{a}+\sqrt{b})^2\right]$, which is strictly wider than the band, the discrepancy being $2\left[\sqrt{ab} - \sqrt{(a-1)(b-1)} - 1\right] \geq 0$ with equality only at $a = b$. So the standard potentials do not even recover the known upper bound. That is not a deficiency of the analysis but of the hypotheses, and it can be seen exactly. Take $A_k = (b/p)I_p$: these are positive semidefinite with $\operatorname{tr} A_k = b$ and $\sum_k A_k = (qb/p)I = aI$, so they satisfy every hypothesis the barrier method uses, and $\mu[A_1, \ldots, A_q]$ is a Laguerre polynomial whose roots converge to the Marchenko–Pastur edges. At $(a,b) = (3,2)$ the upper edge is already exceeded at $p = 4$, where the largest root is 6.229 against $(s+t)^2 = 5.828$, and the lower edge at $p = 48$, where the smallest is 0.154 against $(s-t)^2 = 0.172$. The Marchenko–Pastur interval is therefore attained on those hypotheses, and *no* barrier argument that does not see $\operatorname{rank} A_k \leq b$ can reach the tree band at either edge. Numerically the rank condition is what carries the statement: dropping idempotency but keeping $\operatorname{rank} A_k \leq b$ and $\operatorname{tr} A_k = b$ produced no violation in search, which suggests that class, rather than the projections, as the natural target. What is missing is a barrier built from the Kesten measure of the (a,b)-biregular tree rather than from Marchenko–Pastur, which is to say one that uses that the P_k are 0/1 diagonal — exactly the combinatorial information free probability discards.

That raises the question of how much of the hypothesis is really needed, and the answer delimits the problem usefully. Asking for the same conclusion from *arbitrary* positive semidefinite A_k of rank b with $\sum_k A_k = aI_p$ is false. Take $p=2$, $q=3$, $(a,b)=(3,2)$, and $A_1 = A_2 = \frac{u}{2}I_2$, $A_3 = (3-u)I_2$: each has rank $2 = b$ and the sum is $3I = aI$, while $\mu[A_1,A_2,A_3](y) = y^2 - 6y + \left(6u - \frac{3}{2}u^2\right)$. At $u = \frac{1}{10}$ the constant term is $\frac{117}{200}$ and the roots are 0.09914 and 5.90086, outside the band $[3-2\sqrt{2}, 3+2\sqrt{2}] = [0.17157, 5.82843]$ at both ends; as $u \to 0$ the roots tend to 0 and 6. The counterexample is diagonal, hence commuting, so noncommutativity is not what fails. What fails is that the traces are unconstrained: $\operatorname{tr} A_3 = 5.8 \gg b$. Imposing $\operatorname{tr} A_k = b$ restores the second coefficient, since it then depends on the family only through $\sum_k \|A_k\|_F^2$, which by Cauchy–Schwarz is at least b with equality exactly for idempotents. Projections are thus the extreme point already at the first nontrivial coefficient.

For rank-b orthogonal projections the statement survives, once one adds the hypothesis $a \geq b$, equivalently $p \leq q$, which the graph setting supplies automatically. Without it the statement is false for a trivial reason: if $p > q$ then y^{p-q} divides $\mu[P_1,\ldots,P_q]$, so 0 is a root while the lower band edge is positive. Taking $p=6$, $q=4$, $(a,b)=(2,3)$ with $P_1 = P_2$ and $P_3 = P_4$ the projections onto two complementary 3-spaces gives a double root at 0 against a lower edge of $(\sqrt{2}-1)^2$. A search over more than 10^5 feasible families — real and complex projections, and the intermediate class rank $\leq b$, $\operatorname{tr} = b$ — produced no violation, the smallest observed margin being 3.6% of the band width. That figure needs a caveat, though, and it cuts against the aggregate impression: every tight case in those searches has $b=2$, hence lies in the regime settled by Proposition 32. In the genuinely open range $b \geq 3$ the observed margins are wide — at $(p,q,a,b) = (6,8,4,3)$ the least root is 0.486 against a band edge of 0.101, a factor of nearly five. The numerical evidence for the open part of the statement is thinner than the total counts suggest. Two features make this the right generalization to record. First, commuting rank-b projections with $\sum_k P_k = aI$ are simultaneously diagonalizable and therefore 0/1 diagonal, so they are *exactly* the biregular incidence families: the projection statement is precisely the noncommutative extension of Conjecture 10. Second, the universality of Theorem 13 persists in that setting — for projections the coefficients $E_0,\ldots,E_3$ depend only on (p,q,a,b), because idempotency collapses every trace of a word of length at most three with a repeated letter, while at length four $\operatorname{tr}(A_iA_jA_iA_k)$ does not reduce. The first free invariant sits at level four on both sides of the analogy, matching the 4-cycle count of Theorem 14.

10.2 A determinantal representation

The mixed characteristic polynomial admits a probabilistic representation which gives an unconditional bound with no barrier or stability input. Write $P_k = W_kW_k^*$ with $W_k^*W_k = I_b$, set $U = [W_1|\cdots|W_q]/\sqrt{a}$, and let $\Pi = U^*U$, a rank-p orthogonal projection on $\mathbb{R}^n$, $n = pa = qb$, whose q diagonal $b \times b$ blocks are all $\frac{1}{a}I_b$. Let S be a sample from the determinantal process with kernel Π and $s_k(S)$ the number of slots of block k it uses. Then

$$y^{\,q-p}\,\mu[P_1,\dots,P_q](y) = \mathbb{E}_S \prod_{k=1}^{q} \big(y - a\, s_k(S)\big), \tag{10}$$

by expanding $\det(yI + \sum_k z_k P_k)$ over subsets and applying $\prod_k (1 - \partial_{z_k})$, which annihilates every term using a block twice. Here $|S| = p$ always, so $\sum_k s_k = p$, and each s_k is exactly $\mathrm{Binomial}(b, 1/a)$; for a diagonal family S is simply an independent uniform choice of one incident edge at each vertex of the p-side.

Since $0 \le s_k \le b$, every factor in (10) is positive once $y > ab$, so the expectation is, and there is no root there.

Proposition 19. *Every root of $\mu[P_1,\dots,P_q]$ is at most ab. Moreover, with $m = (a-1)(b-1)$,*

$$ab - \big(\sqrt{a-1} + \sqrt{b-1}\big)^2 = \big(\sqrt{m} - 1\big)^2.$$

So the bound overshoots the conjectured edge by exactly $(\sqrt{m} - 1)^2$, and is lossless precisely when $m = 1$. At $a = b = 2$ the overshoot vanishes and $ab = 4$ is the upper edge; the roots are nonnegative because the matching numbers are, so **the band is proved outright at** $a = b = 2$.

Where ab stands relative to what is known needs care, and we record it exactly. Xu, Xu and Zhu [2, Thm. 1.9] prove, for positive semidefinite $X_1,\dots,X_q$ of rank at most k with $\sum X_i \preceq I$ and $\varepsilon = \max \operatorname{tr} X_i$, that $\operatorname{maxroot} \mu \le \big(\sqrt{1 - \varepsilon/(k-1)} + \sqrt{\varepsilon}\big)^2$ whenever $\varepsilon \le (k-1)^2/k$. Normalizing a rank-$b$ projection family with $\sum_k P_k = aI$ by a puts it in exactly that form with $k = b$ and $\varepsilon = b/a$, and since the roots scale with the family this gives

$$\lambda_{\max} \;\le\; a\Big(\sqrt{1 - \tfrac{b}{a(b-1)}} + \sqrt{\tfrac{b}{a}}\Big)^2, \qquad a \;\ge\; \frac{b^2}{(b-1)^2}, \tag{11}$$

which at $b = 2$ is $a + 2\sqrt{2a-4}$ for $a \ge 4$. That is strictly better than ab for every $a > 4$ — indeed $a + 2\sqrt{2a-4} \le 2a$ with equality only at $a = 4$, formalized as `xxz_le_ab` and `xxz_eq_ab_iff` — and better than Marchenko–Pastur throughout. So ab is the best available bound only where the hypothesis of (11) fails, which at $b = 2$ means $a < 4$: at $a = b = 2$, where it is sharp and settles the band, and at $(a, b) = (3, 2)$, where it gives 6 against a Marchenko–Pastur edge of 9.899 and a band edge of 5.828. Elsewhere (11) is the state of the art:

(a,b)	(11)	ab	Marchenko–Pastur	$a+2\sqrt{a}$	band
$(3,2)$	n/a	6.000	9.899	6.464	5.828
$(4,2)$	8.000	8.000	11.657	8.000	7.464
$(9,2)$	16.483	18.000	19.485	15.000	14.657
$(100,2)$	128.000	200.000	130.284	120.000	119.900
$(4,3)$	10.977	12.000	13.928	–	9.899
$(25,4)$	47.126	100.000	49.000	–	43.971

The comparison also says what the target of this section is worth: at $b = 2$ the bound $a + 2\sqrt{a}$ of Proposition 23 would improve on (11) for every $a > 4$, since $\sqrt{a} \le \sqrt{2a-4}$ there (`target_le_xxz`). It is therefore not a weaker target than the literature already provides, but a stronger one everywhere the barrier hypothesis holds. Proposition 19 is formalized in `RamaLean/ProductBound.lean` as `no_root_above` and `edge_gap`.

The representation also supports an exact recursion on the underlying kernels. Writing N_K for the transversal expansion attached to a kernel K, one has the border identity

$$a\, N_{M+ww^{\mathsf{T}}} = (a-y)\, N_M + N_{\left(\begin{smallmatrix} 0 & w^{\mathsf{T}} \\ w & M \end{smallmatrix}\right)},$$

and Schur complements of admissible kernels remain admissible, with $\delta_e \le 1 - c_e$ forced by $K^2 \preceq K$. This makes an induction on kernel size available in principle. It does not close the upper edge: the scalar cavity invariants that carry the classical argument on graphs are false on the kernel class, witnessed by explicit rational kernels on which the graph unfolding identity fails as an inequality by a y-independent amount. What survives is a reduction of the upper edge on the whole class to a single inequality on bordered kernels, proved for at most two blocks at $b = 2$ by monotonicity of transversal determinants under downdates, which follows from positivity of the adjugate of a positive semidefinite matrix.

10.3 The case of rank two

At $b = 2$ the representation (10) collapses further. Only $s_k \in \{0,1,2\}$ occurs, and $\sum_k s_k = p$ forces the counts of 0s and 1s to be determined by $n_2 = \#\{k : s_k = 2\}$, so with $\Psi(u) = \mathbb{E}\, u^{n_2}$ one has

$$\mu(y) = (y-a)^p\, \Psi\big(\theta(y)\big), \qquad \theta(y) = 1 - \frac{a^2}{(y-a)^2}.$$

Real-rootedness of μ forces Ψ to be real-rooted with nonpositive roots, hence n_2 a sum of independent Bernoulli variables, and the extreme roots of μ are $a\big(1 \pm \sqrt{\pi_{\max}}\big)$ for $\pi_{\max}$ the largest of those parameters. Since the band at $b = 2$ is $[a - 2\sqrt{a-1},\, a + 2\sqrt{a-1}]$, both edges say $a\sqrt{\pi_{\max}} \le 2\sqrt{a-1}$:

Proposition 20. *For $b = 2$, Conjecture 10 for the projection family is equivalent to*

$$\pi_{\max} \;\le\; \frac{4(a-1)}{a^2} \;=\; 1 - \Big(\frac{a-2}{a}\Big)^2.$$

The equivalence of the two edges with each other and with this inequality is formalized in `RamaLean/ProductBound.lean` as `band_iff_pi`; that they coincide is Proposition 32 seen through θ, which is two-to-one about $y = a$.

10.4 The matching form

Recentring at a turns μ into a matching polynomial, and the coefficients have a basis-free form. They also have a Grassmann form: writing $A_k = b_{k1}b_{k1}^{\mathsf{T}} + b_{k2}b_{k2}^{\mathsf{T}}$ and $\omega_k = b_{k1} \wedge b_{k2} \in \Lambda^2\mathbb{R}^p$, the Cauchy–Binet formula gives $e_{2r}\big(\sum_{k\in T} A_k\big) = \big\|\bigwedge_{k\in T}\omega_k\big\|^2$, so

$$\mu(x+a) = \sum_{T\subseteq[q]} (-1)^{|T|} \Big\| \bigwedge_{k\in T} \omega_k \Big\|^2 x^{\,p-2|T|},$$

and M_r is a sum of squared Plücker norms. In the commuting case each norm is 0 or 1 and one recovers the matching numbers.

Theorem 21. *Let $P_1, \dots, P_q$ be rank-two orthogonal projections on $\mathbb{R}^p$ with $\sum_k P_k = aI$. Then*

$$\mu(x+a) = \sum_{r\ge 0} (-1)^r M_r \, x^{p-2r}, \qquad M_r = \sum_{|T|=r} e_{2r}\Big(\sum_{k\in T} P_k\Big),$$

where e_m denotes the m-th elementary symmetric function of the spectrum. The M_r are nonnegative, $M_0 = 1$, $M_1 = q$, and $M_r \le \binom{q}{r}$.

The expression for M_r involves no dilation and no choice of basis for the ranges of the P_k. When the P_k commute, $\sum_{k\in T} P_k$ is a 0/1 diagonal matrix of rank $2r$ exactly when T is a matching of the multigraph whose edges are the blocks, and has rank below $2r$ otherwise; so e_{2r} is the indicator of that event and M_r is the number of r-matchings. Thus at $b = 2$ the recentred mixed characteristic polynomial *is* the matching polynomial in the commuting case, and M_r is its natural noncommutative extension. Verified exactly on $S(K_4)$, $S(K_{3,3})$, the cube, and on noncommuting families including the icosahedral one, where the commutators have norm 0.57.

Two consequences change how the remaining gap should be described. First, in the commuting case Proposition 20 *is* Heilmann–Lieb, so it is a theorem there, and it is sharp: the extremal families are the large-girth a-regular graphs, for which the ratio of $\pi_{\max}$ to its bound rises monotonically through 0.866 at $p = 10$ to 0.949 at $p = 38$. No argument with slack can close it. Second, and more to the point, in that formulation Proposition 20 says exactly that the roots of a mixed characteristic polynomial lie inside the support of the corresponding free convolution, which is the known open sharpening of the Marcus–Spielman–Srivastava bound. So $b = 2$ is not an easier case awaiting a short argument; it is that open problem in another notation.

10.5 The plane class, and two unconditional bounds

What the problem is, at $b = 2$, can be said exactly. Both $\mu(x+a)$ and the operator $\mathrm{Adj}(A) := \sum_k e_2(A_k) P_{\mathrm{range}\, A_k}$ depend on the family only through the pairs $\big(e_2(A_k), \mathrm{range}\, A_k\big)$. So the class is the class of weighted 2-plane families $\{(c_k, V_k) : c_k \ge 0,\ \sum_k c_k P_{V_k} \preceq aI\}$, and $\mu(x+a)$ is the matching polynomial of such a family; for coordinate planes it is the ordinary weighted matching polynomial. The band being sought is therefore the sharp weighted Heilmann–Lieb constant for plane families, which is what fixes $2\sqrt{a}$ as the natural ceiling of any argument of this kind.

In that language the vertex step has an exact form. Setting $\Theta^{(r)} := \sum_{|T|=r} M_{\omega_T} \succeq 0$ and $W(x) := \sum_{r\ge 1} (-1)^r \Theta^{(r)} x^{m-2r}$, one has $\langle e, W(x) e\rangle = F_A(x) - x\, F_{A^{(e)}}(x)$ for every unit vector e, where $F_A(x) = \mu(x+a)$ and $A^{(e)}$ is the compression to $e^\perp$; consequently $\operatorname{tr} W = mF_A - xF'_A$ and $F'_A = \sum_i F_{A^{(e_i)}}$ for any orthonormal basis. The cavity hypothesis is then equivalent to a single matrix inequality: when $\mathrm{Adj}(A) = aI$, the band $[a - 2\sqrt{a},\ a + 2\sqrt{a}]$ holds if and only if

$$W(2\sqrt{a}) + F_A(2\sqrt{a})\, I \ \succeq\ 0,$$

a coordinate-free statement with no recursion in it.

The matrix W has a closed form. If $\omega_T = c_1 \wedge \cdots \wedge c_{2r}$ is decomposable with factor matrix $C = [c_1 | \cdots | c_{2r}]$ and Gram matrix $G = C^{\mathsf{T}} C$, then $\langle \omega_{\hat{j}}, \omega_{\hat{l}}\rangle$ is the (l, j) cofactor of G, so

$$M_{\omega_T} = C \ \mathrm{adj}(G)\, C^{\mathsf{T}}, \tag{12}$$

whence $\operatorname{tr} M_{\omega_T} = 2r\|\omega_T\|^2$ and $\operatorname{tr}\Theta^{(r)} = 2rM_r$, which is the identity $\operatorname{tr} W = mF_A - xF'_A$ again. Two of the levels are forced to be scalar: $\Theta^{(1)} = \operatorname{Adj}(A) = aI$ by hypothesis, and when m is even $\Theta^{(m/2)} = M_{m/2}\, I$, since a top-degree form is a multiple of the volume form and $\|\iota_v\omega\|^2 = \|\omega\|^2\|v\|^2$ for such. Hence for $m \le 4$ every level is scalar, $W(x)$ is a scalar matrix, and the matrix inequality degenerates to a scalar one; from $m = 5$ on it does not, and that is where the content begins.

Behind the matrix inequality there is a vertex recursion valid in every direction, which the graph case sees only along the coordinate axes.

Theorem 22. *Let A be a weighted 2-plane family on $\mathbb{R}^m$ with bivectors ω_k, and let e be a unit vector. Put $f_k = \iota_e\omega_k$, so that*

$$\omega_k = e \wedge f_k + \omega'_k, \qquad f_k,\ \omega'_k \in e^\perp,$$

and ω'_k is the bivector of the compressed family $A^{(e)}$. Then for every $r \ge 1$

$$M_r(A) = M_r\big(A^{(e)}\big) + \sum_{|T|=r} \Big\|\sum_{k\in T} f_k \wedge \omega'_{T\setminus k}\Big\|^2,$$

and consequently, with $N_r(e)$ denoting that sum of squares,

$$F_A(x) = x\,F_{A^{(e)}}(x) - N_e(x), \qquad N_e(x) = \sum_{r\ge 1}(-1)^{r-1}N_r(e)\,x^{m-2r}.$$

Proof. Since e is a unit vector, $\iota_e(\omega_k - e\wedge f_k) = f_k - f_k = 0$, so $\omega_k - e \wedge f_k$ lies in $\Lambda^2(e^\perp)$; writing $b_i = \beta_i e + b'_i$ identifies it as $b'_1 \wedge b'_2$, the bivector of the compressed block. As $e \wedge f_k$ is a 2-form it commutes with the other factors, and $e \wedge e = 0$ kills every term using two of them, so $\omega_T = \omega'_T + e \wedge \sum_{k\in T} f_k \wedge \omega'_{T\setminus k}$. The two summands lie in $\Lambda^{2r}(e^\perp)$ and $e \wedge \Lambda^{2r-1}(e^\perp)$, which are orthogonal, so the norms add; summing over $|T| = r$ gives the first display, and the second is that display multiplied by $(-1)^r x^{m-2r}$ and summed. □

We have looked for Theorem 22 in the literature and not found it. The whole arXiv corpus mentioning “mixed characteristic polynomial” is seven papers, all working through the barrier function or interlacing families; the generalizations of Heilmann–Lieb we can find are to hypergraphs [1], to spanning subgraphs with degree constraints, and to multivariate independence polynomials, all of them graph-theoretic; and a search of arXiv for “matching polynomial” together with any of Grassmann, Plücker, bivector or exterior algebra returns nothing. That is weak evidence: the Gram form below is elementary enough to be folklore, the pre-arXiv matching-polynomial literature is not covered by such a search, and the statement could well appear under different language — hyperbolic or Lorentzian polynomials, or the theory of stable polynomials, where Plücker coordinates do occur but in the different role of characterizing stability by total nonnegativity. We claim only that we have not found it.

The exterior algebra can be dispensed with entirely, and doing so identifies the theorem with a known lemma. Let C be the $m \times 2r$ matrix whose columns are the factors of ω_T, so that $\|\omega_T\|^2 = \det(C^\mathsf{T} C)$. Splitting off e is the decomposition $C = eg^\mathsf{T} + C'$ with $g = C^\mathsf{T} e$ and $C' = (I - ee^\mathsf{T})C$ the compressed factor matrix; since $e^\mathsf{T} C' = 0$ and $\|e\| = 1$,

$$C^\mathsf{T} C = C'^\mathsf{T} C' + gg^\mathsf{T}.$$

So the content of Theorem 22 is the *matrix determinant lemma*

$$\det\bigl(A + gg^{\mathsf T}\bigr) = \det A + g^{\mathsf T}\operatorname{adj}(A)\, g, \qquad A = C'^{\mathsf T} C', \tag{13}$$

and the nonnegativity of the cavity term is nothing but positivity of the adjugate of a positive semidefinite matrix. In particular $M_r(A) \ge M_r(A^{(e)})$ needs no computation at all. Splitting g into its r blocks of two identifies the pieces: the block-diagonal part of the quadratic form (13) is $\sum_k \theta_k M''_{k,r-1}$ and the off-block-diagonal part is C_r, which is how the cross terms should be read.

For a coordinate family at a coordinate direction $e = e_v$ one has $f_k = \pm e_u$ for the edges $k = uv$ and $f_k = 0$ otherwise, the cross terms vanish, and $N_r(e_v) = \sum_{u\sim v} m_{r-1}(G-v-u)$: the theorem is then the Heilmann–Lieb vertex recursion. At a generic direction, or for a family that is not diagonal, the cross terms do not vanish and the recursion is not the classical one. Two things follow at once. First, taking r arbitrary, $M_r(A) \ge M_r(A^{(e)})$ for every unit e: the matching numbers of a plane family decrease under compression in every direction, which for graphs is $m_r(G) \ge m_r(G-v)$. Second, the whole band reduces to the sign of the cross terms.

Proposition 23. *Write $X_e := N_e - \sum_k \theta_k F_{A''_k}$ for the cross-term part of N_e, where $\theta_k = \|f_k\|^2$ and A''_k is the compression to $\{e, f_k\}^{\perp}$. If $X_e(x) \le 0$ for every weighted 2-plane family with $\operatorname{Adj} \preceq aI$, every unit e and every $x \ge 2\sqrt{a}$, then every such family satisfies the band $[a - 2\sqrt{a},\, a + 2\sqrt{a}]$.*

Proof. Induct on m. Put $R_e = F_A/F_{A^{(e)}}$ and $R'_k = F_{A^{(e)}}/F_{A''_k}$. Dividing Theorem 22 by $F_{A^{(e)}}$,

$$R_e = x - \sum_k \frac{\theta_k}{R'_k} - \frac{X_e}{F_{A^{(e)}}}.$$

The class is closed under compression, so the inductive hypothesis $R'_k \ge x/2$ applies, and $\sum_k \theta_k = \sum_k \|\iota_e \omega_k\|^2 = \langle e, \operatorname{Adj}(A) e\rangle \le a$. Hence $R_e \ge x - 2a/x$, and $x - 2a/x \ge x/2$ exactly when $x \ge 2\sqrt{a}$. The base case $m \le 1$ is $R_e = x$. So $R_e \ge x/2 > 0$ for $x \ge 2\sqrt{a}$, and $F_A = R_e F_{A^{(e)}}$ has no root there; F_A is even or odd, so all roots lie in $[-2\sqrt{a}, 2\sqrt{a}]$. $\square$

The leading cross term can be evaluated exactly, and it is a square.

Theorem 24. *Let A be a weighted 2-plane family on $\mathbb{R}^m$ and e a unit vector, and put $f_k = \iota_e \omega_k$, $\omega'_k = \omega_k - e \wedge f_k$ and $u_k = \iota_{f_k} \omega'_k$. Then*

$$C_2 = \Bigl\|\sum_k f_k \otimes \omega'_k\Bigr\|^2 - \Bigl\|\sum_k u_k\Bigr\|^2,$$

and $\sum_k u_k$ is the off-diagonal $(e, e^{\perp})$ block of $\operatorname{Adj}(A)$. So if $\operatorname{Adj}(A) = aI$ then

$$C_2 = \Bigl\|\sum_k f_k \otimes \omega'_k\Bigr\|^2 \;\ge\; 0,$$

with equality if and only if $\sum_k f_k \otimes \omega'_k = 0$.

Proof. For a vector u and forms α, γ one has $\iota_w(u \wedge \alpha) = \langle w, u\rangle\alpha - u \wedge \iota_w\alpha$, hence

$$\langle u \wedge \alpha,\ w \wedge \gamma\rangle = \langle u, w\rangle\langle \alpha, \gamma\rangle - \langle \iota_w\alpha, \iota_u\gamma\rangle. \tag{14}$$

Applying (14) to each cross term with $\alpha = \omega'_l$, $\gamma = \omega'_k$ gives $C_2 = \sum_{k\neq l}\big[\langle f_k, f_l\rangle\langle \omega'_k, \omega'_l\rangle - \langle u_k, u_l\rangle\big]$.

Now $f_k = \iota_e\omega_k$ lies in the plane of ω_k, and it is orthogonal to e, so it lies in the image of that plane under the projection to $e^\perp$ — which is the plane of ω'_k. Hence $f_k \wedge \omega'_k = 0$, and since $\|f_k\|^2\|\omega'_k\|^2 = \|\iota_{f_k}\omega'_k\|^2 + \|f_k \wedge \omega'_k\|^2$ for simple ω'_k, this says exactly $\|f_k\|^2\|\omega'_k\|^2 = \|u_k\|^2$. Therefore

$$\sum_{k\neq l}\langle f_k, f_l\rangle\langle \omega'_k, \omega'_l\rangle = \Big\|\sum_k f_k \otimes \omega'_k\Big\|^2 - \sum_k \|u_k\|^2, \qquad \sum_{k\neq l}\langle u_k, u_l\rangle = \Big\|\sum_k u_k\Big\|^2 - \sum_k \|u_k\|^2,$$

and subtracting gives the display. For the last claim, polarizing $\langle v, \Theta_k v\rangle = \|\iota_v\omega_k\|^2$ at $v = e$ against $v \in e^\perp$ gives $\langle e, \mathrm{Adj}(A)v\rangle = -\langle v, \sum_k u_k\rangle$, so $\mathrm{Adj}(A) = aI$ forces $\sum_k u_k = 0$. □

Theorem 24 is sharp in the way the data show: over coordinate families $\min_e C_2 = 0$ to machine precision, attained exactly at the coordinate directions, where each summand $f_k \otimes \omega'_k$ vanishes separately — every ω_k either contains e in its plane, so $\omega'_k = 0$, or is orthogonal to it, so $f_k = 0$. That is precisely why the classical recursion has no cross terms. Off that locus the minimum is strictly positive.

The proof reindexes, and doing so gives every level at once.

Theorem 25. *For every $r \geq 2$,*

$$C_r = \sum_{|S|=r-2}\Big[\,Q_S - \|W_S\|^2\,\Big], \qquad \begin{aligned} Q_S &= \sum_{k,l\notin S}\langle f_k, f_l\rangle\langle \omega'_k \wedge \omega'_S,\ \omega'_l \wedge \omega'_S\rangle,\\ W_S &= \sum_{k\notin S}\iota_{f_k}\big(\omega'_k \wedge \omega'_S\big),\end{aligned}$$

and $Q_S = \big\|\sum_{k\notin S} f_k \otimes (\omega'_k \wedge \omega'_S)\big\|^2 \geq 0$.

Proof. Writing $T = S \cup \{k, l\}$ reindexes C_r as $\sum_{|S|=r-2}\sum_{k\neq l\notin S}\langle f_k \wedge \omega'_l \wedge \omega'_S,\ f_l \wedge \omega'_k \wedge \omega'_S\rangle$. Apply (14) with $\alpha = \omega'_l \wedge \omega'_S$ and $\gamma = \omega'_k \wedge \omega'_S$. The wedge $\omega'_k \wedge \omega'_S$ is a product of vectors, hence simple, and $f_k \wedge (\omega'_k \wedge \omega'_S) = (f_k \wedge \omega'_k) \wedge \omega'_S = 0$ by the proof of Theorem 24; so $\|f_k\|^2\|\omega'_k \wedge \omega'_S\|^2 = \|\iota_{f_k}(\omega'_k \wedge \omega'_S)\|^2$, and the diagonal terms of the two double sums cancel exactly as before. The Gram matrix of the vectors $\omega'_k \wedge \omega'_S$ is positive semidefinite, so Q_S is the squared norm of $\sum_{k\notin S} f_k \otimes (\omega'_k \wedge \omega'_S)$. □

At $r = 2$ the set S is empty, $W_\emptyset = \sum_k \iota_{f_k}\omega'_k$ vanishes by tightness, and Theorem 25 is Theorem 24. At $r = 3$ it does not vanish, but it simplifies: since $\iota_{f_n}\omega'_n$ lies in the plane of ω'_n, the self term $(\iota_{f_n}\omega'_n) \wedge \omega'_n$ is zero, and using $\sum_k u_k = 0$,

$$W_{\{n\}} = \sum_{k\neq n}\omega'_k \wedge \iota_{f_k}\omega'_n.$$

So the obstruction at $r = 3$ is exactly the failure of the compressed planes to be mutually orthogonal, and it is the only obstruction: everything else is a square.

It is worth being exact about what this settles. Writing the condition $X_e \le 0$ at $x = 2\sqrt{a}$ in the form

$$\sum_{r\ge 2}(-1)^r C_r\,(4a)^{2-r} \;\ge\; 0, \tag{15}$$

Theorem 24 is the leading coefficient of (15), that is the case $a \to \infty$. It gives $X_e \le 0$ outright when $m \le 5$, where C_2 is the only cross term. That range does not improve on the two-moment bounds above, so Theorem 24 does not by itself enlarge the unconditional range; what it supplies is the first structural reason for the sign.

The obvious next step, bounding C_3 by $4a\,C_2$ so as to reach $m \le 7$, is not available, and it fails for a reason worth recording rather than for lack of effort. The ratio $4a\,C_2/C_3$ decreases steadily in m — over cubic graphs it runs 3.49 at $m = 8$, 2.00 at $m = 10$ (Petersen), 1.38 at $m = 12$ (Franklin), 1.03 at $m = 14$ (Heawood) — and crosses below 1 at $m = 16$, where the Möbius–Kantor graph gives 0.827, then 0.687 (Pappus, $m = 18$), 0.590 (Desargues, $m = 20$) and 0.457 (Nauru, $m = 24$). So $C_3 \le 4aC_2$ is false in general, and increasingly so.

The reason is visible one level down. For odd r the criterion wants C_r small, that is $\|W_S\|^2$ close to Q_S; but the ratio $\sum_n \|W_n\|^2/\sum_n Q_n$ decays, running 0.207, 0.190, 0.154, 0.119, 0.101 over the same graphs at $m = 8, \dots, 16$. So C_3 is essentially $\sum_n Q_n$, an undamped sum of Schur squares, and the contraction W supplies almost no cancellation. Whatever makes (15) hold is therefore the alternation *between* levels, not any cancellation *within* one.

What the same computation shows is that the truncation is what fails, not the conclusion. Writing Σ_R for the partial sum of (15) over $2 \le r \le R$:

G	m	Σ_2	Σ_3	Σ_4	Σ_5	Σ_6
cube	8	1.268	0.905	0.926		
Petersen	10	1.624	0.812	0.891	0.891	
Franklin	12	1.713	0.468	0.730	0.713	0.713
Heawood	14	2.007	0.060	0.679	0.603	0.607
Möbius–Kantor	16	2.139	−0.448	0.669	0.458	0.476
Pappus	18	2.142	−0.974	0.739	0.291	
Desargues	20	2.235	−1.552	0.969	0.127	
Nauru	24	2.231	−2.656	1.786		

The values are for one direction per graph, fixed by the seed in `code/hl_Cr.py`, which also regenerates Theorem 25 against the definition of C_r. The terms $|C_r|(4a)^{2-r}$ peak at $r = 3$ and then decay geometrically — at $m = 16$ they run 2.14, 2.59, 1.12, 0.211, 0.0176 — so the series converges, but the number of terms needed grows with m: three nearly suffice at $m = 14$, four at $m = 16$, and at $m = 20$ and 24 the partial sums have not settled by Σ_5. So (15) is not a statement of bounded length, and in particular the two-term reading fails outright from $m = 16$. By Theorem 25 every term is a Schur square minus the square of a contraction, so what a proof must supply is control of the alternating sum of the Q_S across levels.

How much room the inequality actually has is worth measuring, and the answer is that there is a great deal — and that the recursion cannot use any of it. Evaluating the exact ratio $R_e = F_A/F_{A^{(e)}}$ at $x = 2\sqrt{a}$ on regular graphs, in units of the threshold $x/2 = \sqrt{a}$:

a	graphs	$R_e/(x/2)$	spread
3	cube, Petersen, Franklin, Heawood	1.3662–1.3692	$3.0 \cdot 10^{-3}$
4	$C_8(1,2)$, $C_{10}(1,2)$, $C_{12}(1,2)$	1.3488–1.3497	$8.6 \cdot 10^{-4}$
5	$C_8(1,2,4)$, $C_{10}(1,2,5)$	1.3246–1.3322	$7.5 \cdot 10^{-3}$

The ratio is a function of a alone to three digits, independent of the graph and of the dimension, and exceeds the threshold by a third. The slack $-X_e/(F_{A^{(e)}}\sqrt{a})$ likewise *increases* with m, running $0.033, 0.044, 0.049, 0.058$ over the four cubic graphs above.

That constant can be identified exactly. Averaging the vertex identity over directions gives $\mathbb{E}_e F_{A^{(e)}} = F'_A/m$ — which is $F'_A = \sum_i F_{A^{(e_i)}}$ again — so the direction-averaged ratio is

$$\bar{R} = \frac{mF_A}{F'_A} = \Big(\frac{1}{m}\sum_i \frac{1}{x-\rho_i}\Big)^{-1} = \frac{1}{G(x)},$$

G the Stieltjes transform of the root measure of F_A. For a coordinate family F_A is the matching polynomial, and for a-regular graphs of growing girth its root measure converges to the Kesten–McKay measure, whose Stieltjes transform is $G(z) = 2(a-1)/\big((a-2)z + a\sqrt{z^2-4(a-1)}\big)$. At $z = 2\sqrt{a}$ the radical is *exactly* 2, and everything collapses:

Proposition 26. $\dfrac{1}{G(2\sqrt{a})}\Big/\sqrt{a} = \dfrac{\sqrt{a}+2}{\sqrt{a}+1} = 1 + \dfrac{1}{1+\sqrt{a}}.$

So the cavity ratio exceeds the threshold the induction can propagate by exactly $1/(1+\sqrt{a})$, and the excess tends to 0 as a grows — which is the analytic shadow of $2\sqrt{a}$ being asymptotically attained.

The limit appears to be attained from above, never crossed, which suggests that Proposition 26 states an infimum rather than merely a limit:

Conjecture 27. *For every weighted 2-plane family with* $\mathrm{Adj}(A) = aI$ *and every unit e,*

$$\frac{R_e}{x/2} \;\geq\; 1 + \frac{1}{1+\sqrt{a}} \qquad \textit{at } x = 2\sqrt{a}.$$

This was not violated in any family tested, over graphs and over general weighted plane families, minimizing over directions; the smallest margin observed was $2.2 \cdot 10^{-4}$, at the highest-girth cubic graph, which is where Proposition 26 says it should be. Conjecture 27 is strictly stronger than $X_e \leq 0$, which is its $c = 1$ case, and by Proposition 28 it cannot be proved by this induction either — the recursion propagates $c = 1$ and nothing else, whatever is true. Against the exact values the prediction is 1.366025 at $a = 3$, 1.333333 at $a = 4$ and 1.309017 at $a = 5$, and the error decays with the girth exactly as Kesten–McKay convergence requires: over cubic graphs $3.2 \cdot 10^{-3}$ at girth 4 (the cube), $9.2 \cdot 10^{-4}$ at girth 5 (Petersen), and $2.2 \cdot 10^{-4}$, $1.7 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$, $1.2 \cdot 10^{-4}$ at girth 6 (Heawood, Möbius–Kantor, Pappus, Desargues). None of that slack is available to the argument:

Proposition 28. *Fix $a > 0$ and suppose the inductive hypothesis of Proposition 23 is strengthened from $R \geq x/2$ to $R \geq c\,(x/2)$ for some $c > 0$. At $x = 2\sqrt{a}$ the recursion sustains this only for $c = 1$.*

Proof. Propagating $R \geq c\sqrt{a}$ requires $c\sqrt{a} \leq 2\sqrt{a} - a/(c\sqrt{a})$, which after clearing the denominator is $a(c-1)^2 \leq 0$, so $c = 1$. $\square$

There is one hypothesis the argument has not used, and it changes what has to be proved. The class is *not* closed under compression in the tight sense:

Proposition 29. *For a weighted* 2*-plane family with* $\operatorname{Adj}(A) = aI$ *and any unit* e*,*

$$\operatorname{Adj}\big(A^{(e)}\big) = a\, I_{e^\perp} - F, \qquad F = \sum_k f_k f_k^\mathsf{T}, \qquad \operatorname{tr} F = a.$$

Proof. For $u, v \perp e$ one has $\iota_u\omega_k = -\langle u, f_k\rangle e + \iota_u\omega_k'$ with $\iota_u\omega_k' \perp e$, so $\langle \iota_u\omega_k, \iota_v\omega_k\rangle = \langle u, f_k\rangle\langle v, f_k\rangle + \langle \iota_u\omega_k', \iota_v\omega_k'\rangle$; summing and using $\operatorname{Adj}(A) = aI$ gives the display, and $\operatorname{tr} F = \sum_k \theta_k = \langle e, \operatorname{Adj}(A)e\rangle = a$. $\square$

So compressing costs exactly a positive semidefinite matrix of trace a, and the children of a tight family are deficient by a definite amount. Spending that deficiency strengthens the induction. Write $d(e) = a - \langle e, \operatorname{Adj}(A)e\rangle \geq 0$ for the local deficiency, and replace the hypothesis $R \geq x/2$ of Proposition 23 by

$$\text{(H)} \qquad R_e \;\geq\; \frac{x}{2} + \frac{2\, d(e)}{x}.$$

This is consistent at the base ($m \leq 1$: $R_e = x$, $d = a$, and $x/2 + 2a/x = x$ exactly), and at a tight family $d \equiv 0$, so it still delivers the band.

Proposition 30. *Let* $g_k = \langle \hat f_k, F \hat f_k\rangle$ *be the deficiency the children inherit. Then* (H) *propagates provided*

$$X_e \;\leq\; \frac{2F_{A^{(e)}}}{x} \sum_k \frac{\theta_k\, g_k}{a + g_k},$$

and the estimate is lossless: the two sides of the resulting comparison are identically equal.

The right-hand side is *strictly positive*, since $g_k \geq \theta_k > 0$. So the strengthened induction does not require $X_e \leq 0$; it requires X_e below an explicit positive quantity. On the graphs tested (H) held with margin 0.52 to 0.71, and the sufficient condition held with the right-hand side exceeding $|X_e|$ by a factor of five to twenty.

That is worth having because an upper bound on X_e does exist, and the deficiency supplies it. Applying the vertex identity a second time writes each child polynomial as $F_{A_k''} = \big(F_{A^{(e)}} - q_k\big)/x$ with $q_k = \langle \hat f_k, W^{(e)}(x) \hat f_k\rangle$, so $\sum_k \theta_k F_{A_k''} = \big(aF_{A^{(e)}} - \operatorname{tr}(FW^{(e)})\big)/x$, and substituting into $X_e = N_e - \sum_k \theta_k F_{A_k''}$ gives a relation between the vertex matrices at two consecutive levels:

Proposition 31. $x\, X_e = \operatorname{tr}\big(F\, W^{(e)}(x)\big) - x\,\langle e, W(x)e\rangle - a\, F_{A^{(e)}}(x)$*, and consequently*

$$x\, X_e \;\leq\; a\, \lambda_{\max}\big(W^{(e)}(x)\big) - x\,\langle e, W(x)e\rangle - a\, F_{A^{(e)}}(x).$$

Proof. The identity is the substitution above. For the bound, $F = \sum_k \theta_k \hat f_k \hat f_k^\mathsf{T}$ is a nonnegative combination of unit rank-ones with $\sum_k \theta_k = \operatorname{tr} F = a$, so $\operatorname{tr}(FW^{(e)}) = \sum_k \theta_k q_k \leq \big(\sum_k \theta_k\big)\lambda_{\max}(W^{(e)})$; no spectral theory is needed beyond the definition of $\lambda_{\max}$. $\square$

This is the first upper bound on X_e in the development — every earlier statement bounded it below or computed it exactly — and the trace step is not lossy where it matters: on K_5 and K_6, where $X_e = 0$, it holds with equality. Together with Proposition 30, whose target is strictly positive, the band is therefore reduced to a spectral bound on the child's vertex matrix. We have not proved that bound.

The constant is therefore not an artefact of the estimate: $\sqrt{a}$ is the fixed point of $\rho \mapsto x - a/\rho$ at $x = 2\sqrt{a}$, the value at which the two roots of $\rho^2 - x\rho + a$ collide. For $x > 2\sqrt{a}$ the inductive hypothesis has an interval of admissible thresholds and the argument has slack; at $x = 2\sqrt{a}$ the interval is a point and it has none. That is exactly why $2\sqrt{a}$, and not the conjectured $2\sqrt{a-1}$, is the ceiling of every argument of this shape. The cross terms $C_r = N_r - \sum_k \theta_k M''_{k,r-1} = \sum_{|T|=r} \sum_{k \neq l \in T} \langle f_k \wedge \omega'_{T\setminus k}, f_l \wedge \omega'_{T\setminus l} \rangle$ were nonnegative in every family tested, and at $r = 2$ that is now Theorem 24. The nonnegativity is not termwise, and it is worth recording that it is not, since the termwise statement is the natural thing to try and it is false: single pairs (k, l) and even whole summands $\sum_{k \neq l \in T}$ take negative values, the smallest observed being $-1.9 \cdot 10^{-3}$ and $-3.8 \cdot 10^{-3}$ over weighted plane families in dimensions six and eight. If $C_r \geq 0$ holds it is therefore a statement about the sum over all T of a given size, not about its terms. Summed, it survives minimization over the sphere, and sharply: for coordinate families $\min_e C_2 = 0$ to machine precision, attained exactly at the coordinate directions, where the classical recursion lives, while for noncommuting projection and general plane families the minimum is strictly positive, growing from 0.048 at $m = 4$ to 0.777 at $m = 8$. Finally $X_e = \sum_{r \geq 2} (-1)^{r-1} C_r x^{m-2r}$ was nonpositive at $x = 2\sqrt{a}$ in every case under direct maximization over the sphere — zero to $4 \cdot 10^{-12}$ for coordinate families, strictly negative otherwise — over graph families in dimensions $4, 5, 6$, projection families in dimensions $4, 6$ and weighted plane families in dimension 4. The identity of Theorem 22 was checked to 10^{-13} on graph, projection and general weighted plane families at generic directions (`code/hl_Wspec.py`, `code/hl_Xsign.py`).

Two bounds follow unconditionally from real-rootedness alone. Writing $F_A = x^m - M_1 x^{m-2} + M_2 x^{m-4} - \cdots$, one has $M_1 = \frac{1}{2} \operatorname{tr} \operatorname{Adj}(A) \leq am/2$ and, using $\|\omega_j \wedge \omega_l\|^2 \geq 1 - \operatorname{tr}(P_j P_l)$, $M_1^2 - 2M_2 \leq \operatorname{tr} \operatorname{Adj}(A)^2 - \sum_k c_k^2$. Since the largest root λ satisfies $\lambda^2 \leq M_1$ and $\lambda^4 \leq M_1^2 - 2M_2$, the band $[a - 2\sqrt{a}, a + 2\sqrt{a}]$ holds whenever $m \leq 8$, and whenever $m \leq 16 + \left(\sum_k c_k^2\right)/a^2$, which for projection families is $m \leq 32a/(2a-1)$. The second reproduces the known two-moment bound for projections and extends it to all weighted plane families. Adversarial search over rank-two projection families and over random biregular graphs reaches 0.83 of the threshold at $a = 3$ and 0.72 at $a = 4$, with no violation; graphs are marginally more extremal than general projections.

10.6 The reflection for projections

At $b = 2$ the whole question also collapses onto the known half, by a symmetry.

Proposition 32. *Let $P_1, \ldots, P_q$ be rank-2 orthogonal projections on $\mathbb{R}^p$ with $\sum_k P_k = aI$. Then $\mu[P_1, \ldots, P_q](2a - y) = (-1)^p \mu[P_1, \ldots, P_q](y)$; in particular $\lambda_{\min} + \lambda_{\max} = 2a$.*

Proof. The mixed characteristic polynomial is affine-linear in each slot, so it commutes with taking expectations slotwise, and for rank-one slots $\mu[v_1 v_1^\mathsf{T}, \ldots] = \det\left(yI - \sum_k v_k v_k^\mathsf{T}\right)$, since a multi-affine polynomial is sent by $\prod_k (1 - \partial_k)$ to its value at $z = (-1, \ldots, -1)$. Choose an orthonormal pair e_k, f_k spanning the range of P_k and set $v_k = e_k + \varepsilon_k f_k$ with ε_k independent uniform signs. Then $\mathbb{E}\, v_k v_k^\mathsf{T} = P_k$ and $v_k v_k^\mathsf{T} = P_k + \varepsilon_k S_k$ with $S_k = e_k f_k^\mathsf{T} + f_k e_k^\mathsf{T}$ and $S_k^2 = P_k$, so $\mu(y) = \mathbb{E}_\varepsilon \det\big((y-a)I - \sum_k \varepsilon_k S_k\big)$. That is an even or odd function of $t = y - a$ according to the parity of p, because $\varepsilon \mapsto -\varepsilon$ preserves the distribution. $\square$

The last step, which carries the whole proposition, is formalized in `RamaLean/SignReflection.lean` as `sign_avg_reflect`: for any family of matrices $S_1, \ldots, S_q$ the sign average $\sum_{\varepsilon \in \{\pm 1\}^q} \det\big(tI - \sum_k \varepsilon_k S_k\big)$ satisfies $\rho(-t) = (-1)^n \rho(t)$, with no hypothesis on the S_k; the shift and the exchange of band edges $2a - (\sqrt{a-1}+1)^2 = (\sqrt{a-1}-1)^2$ are `mixedChar_reflect` and `band_edges_swap`. The step identifying the sign average with μ is formalized in the same file: by `sum_prod_sgn` a nonempty sign monomial averages to zero over $\{\pm 1\}^q$, proved by the same involution, so by `sum_multiaffine` the average of a multi-affine function returns 2^q times its constant term — which for $\det\big(yI + \sum_k z_k A_k\big)$ with rank-one A_k is what $\prod_k (1 - \partial_{z_k})$ extracts.

Since $2a - (\sqrt{a-1}+1)^2 = (\sqrt{a-1}-1)^2$, the two band edges are exchanged by the reflection, so at $b = 2$ the lower bound is *equivalent* to the upper one. For diagonal families the upper bound is Godsil's path-tree theorem and $b = 2$ is therefore settled — that is Theorem 11 again. For projections it is not: the path tree is a graph construction, and the only bound available in that generality is the Marchenko–Pastur edge $(\sqrt{a} + \sqrt{b})^2$, which is strictly larger than $(s+t)^2$. So at $b = 2$ the projection statement is reduced to its upper half, not closed. The mechanism in the diagonal case is transparent: there $\sum_k \varepsilon_k S_k$ is a signed adjacency matrix, the expectation is Godsil–Gutman, and one recovers $\nu_{S(H)}(y) = \mu_H(y-a)$.

This also locates the obstruction at $b \geq 3$ sharply. Writing $N_k = v_k v_k^\mathsf{T} - P_k$ one has $N_k^2 = (b-2)N_k + (b-1)P_k$, with spectrum $\{b-1, -1^{(b-1)}\}$ on the range of P_k; the reflection needs this invariant under $N_k \mapsto -N_k$, which happens exactly when $b = 2$. Equivalently: the mean root is always a, since $\sum_i \lambda_i = \operatorname{tr} \sum_k P_k = pa$, while the band centre is $a + b - 2$. The conjecture therefore asserts a skew of exactly $b - 2$, which is the operator form of the displacement in Theorem 12 and is invisible to any reflection or first-moment argument.

The source of that skew can be named exactly. The jump N_k has spectrum $\{b - 1, (-1)^{(b-1)}\}$ on the range of P_k, whose moments are

$$m_1 = 0, \qquad m_2 = b(b-1), \qquad m_3 = b(b-1)(b-2).$$

So the third moment vanishes precisely when $b = 2$: Proposition 32 is not an accident of small b but the vanishing of the jump skewness, and for $b \geq 3$ any argument reaching the lower edge must see a third-order quantity.

11 The band is the support of a free convolution power

There is a single reason behind all of this, and it is worth stating because it accounts uniformly for the failure of every method tried. Let $\chi = (1 - \frac{1}{b})\delta_0 + \frac{1}{b}\delta_b$, and let ν be the centred slot spectrum $\frac{1}{b}\delta_{b-1} + (1 - \frac{1}{b})\delta_{-1}$, so that $\chi = \delta_1 \boxplus \nu$.

Proposition 33. $\operatorname{supp}\big(\chi^{\boxplus a}\big) = \big[(\sqrt{a-1} - \sqrt{b-1})^2,\ (\sqrt{a-1} + \sqrt{b-1})^2\big]$.

So the band is the support of a free convolution *power*. Every method that fails computes instead a compound free Poisson of rate a with jump ν, whose free cumulants are $a\, m_n(\nu)$

rather than $a\,\kappa_n(\nu)$, and

$$m_n(\nu) = \kappa_n(\nu) \quad (n \le 3), \qquad \kappa_4(\nu) - m_4(\nu) = -2(b-1)^2.$$

The two sequences agree through third order and part company at the fourth, by exactly $-2a(b-1)^2$. Order four is where every structure in this problem turns over, and it does so three independent times: the coefficients $E_0, \dots, E_3$ are determined by the parameters while E_4 carries the 4-cycle count; the moment and free-cumulant sequences of the jump agree up to order three and split at order four; and the parallel-class construction for the finite free convolution reproduces the elementary symmetric functions e_1, e_2, e_3 of μ and first fails at e_4. That is why the barrier bound, the scalar free convolution and the finite free convolution all overshoot by the same margin of about 1.1, and it is the analytic form of the third-moment obstruction above.

The finite free cumulants of μ itself follow the free power, not the Poisson:

$$\kappa_1 = a, \qquad \kappa_2 = \frac{ap(b-1)}{p-1}, \qquad \kappa_3 = \frac{ap^2(b-1)(b-2)}{(p-1)(p-2)},$$

with κ_4 the first that depends on the family — matching the coefficient ladder, where $E_0, \dots, E_3$ are universal and E_4 carries the 4-cycle count. Note $\kappa_3 = 0$ exactly when $b = 2$: Proposition 32 is visible in the cumulants.

Two further facts delimit what can be done with this. Writing $\psi_0(x) = x^{p-p/b}(x-b)^{p/b}$ and $\Psi = \psi_0^{\boxplus_p a}$, the polynomial Ψ satisfies the band at every p, but $\mu \ne \Psi$ in general and there is no real-rooted ρ with $\mu = \Psi \boxplus_p \rho$ unless the two agree: finite free deconvolution is unique, so such a ρ would have $\kappa_1 = \kappa_2 = 0$, and κ_2 is a positive multiple of the variance of the roots, forcing $\rho = x^p$. The very agreement of the first three cumulants is what forbids the factorization. Divisibility by a single copy, $\mu = \psi_0 \boxplus_p \rho$ with ρ real-rooted, would by the Marcus–Spielman–Srivastava bound give a lower bound on $\lambda_{\min}$, and it holds in small cases; but it is not a property of the class. Taking $b = 2$ and the incidence families of 3-regular graphs, the deconvolution ρ is real-rooted for $S(K_4)$ at $p = 4$ and for $K_{3,3}$ at $p = 6$, and fails for the cube Q_3 at $p = 8$ and the Petersen graph at $p = 10$, where the largest imaginary part of a root of ρ is 0.133 and 0.290 respectively. The cube is the informative case: it is 3-regular and bipartite, hence 1-factorable, so the failure is not caused by the absence of a partition of the blocks into parallel classes. Whatever separates the projection class from the positive semidefinite relaxation, single-copy divisibility is not it: at $(p, a, b) = (12, 4, 2)$ the scalar family, which violates the band, admits the factorization while a projection family does not.

Two general facts about free convolution came out of the attempt, and they bound what any argument of this shape can achieve. For finitely supported probability measures,

$$\min\operatorname{supp}\omega + \min\operatorname{supp}\tau \;\le\; \min\operatorname{supp}(\omega \boxplus \tau) \;\le\; \min\operatorname{supp}\tau + m_1(\omega), \tag{16}$$

the lower bound from the operator model and the upper from $\min\operatorname{supp}(\omega \boxplus \tau) = \sup_{w<0}[K_\tau(w) + R_\omega(w)]$ together with the monotonicity of R_ω, which follows from Cauchy–Schwarz. And if $\min\operatorname{supp}\tau \ge c$ then $\tau(\{c\}) \le \mu_2/\big(\mu_2 + (m_1 - c)^2\big)$, so by Bercovici–Voiculescu an atom of mass exceeding $\omega(\{\min\operatorname{supp}\omega\})$ pins the left edge of $\omega \boxplus \tau$ at c exactly.

Applied with $\omega = \chi$, for which $m_1 = 1$, (16) says that convolving with χ moves the left edge by at most 1. Any certificate that tries to reach the band edge by exhibiting μ as a free convolution with χ is therefore capped, and the first three cumulants do not constrain the other factor enough: there are measures with exactly the forced values of $\kappa_1, \kappa_2, \kappa_3$, real-rooted, supported in $[0, ab]$, whose convolution with χ has left edge 0. At $(a, b) = (5, 4)$, $p = 12$ one may take $x^4(x-4)^3(x-6)^4(x-12)$, whose mass at 0 exceeds $1/b$.

The same atom mechanism explains the hypothesis $a \geq b$: for $a < b$ the measure $\chi^{\boxplus a}$ carries an atom at 0 of mass $1 - a/b$, so its left edge is 0 and the band statement is empty.

12 Rank monotonicity, and what it explains

What does separate them can be said exactly, in terms of the representation (10). For a general positive semidefinite family with $\sum_k A_k = aI$ the marginal of s_k has probability generating function $\prod_i \bigl(\frac{a-\alpha_i}{a} + \frac{\alpha_i}{a} z\bigr)$ over the eigenvalues α_i of A_k, a Poisson binomial; it is exactly $\mathrm{Binomial}(b, 1/a)$ if and only if every α_i lies in $\{0, 1\}$, that is if and only if A_k is a rank-b projection. Among Poisson binomials of fixed mean b/a with n equal parameters the variance is $(b/a)(1-b/(na))$, increasing in n; so the projections, where $n = b$, are the *variance-minimizing* extreme point of the whole family. The scalar counterexample $A_k = (b/p)I$ sits at the opposite end, at the Poisson limit, with variance exceeding the projection value by exactly b/a^2.

This has a consequence that organizes the whole picture. Fix (a, b) and let the common rank n of the A_k vary, with $A_k = (b/n)Q_k$ for rank-n projections Q_k summing to $(an/b)I$, so that $\operatorname{tr} A_k = b$ and $\sum_k A_k = aI$ throughout. At $(a, b) = (4, 2)$, $p = 6$, $q = 12$ the extreme roots move monotonically:

n	least root	greatest root
2	1.129	6.871
3	1.004	7.706
4	0.951	8.090
5	0.921	8.317
6	0.903	8.467

against the band $[0.536, 7.464]$ at $n = b = 2$ and the Marchenko–Pastur interval $[0.343, 11.657]$ approached as $n \to p$. So the conjectured band is the endpoint at $n = b$ of a family of intervals increasing in n, whose other endpoint is Marchenko–Pastur, and the Marchenko–Pastur value is attained. That is the correct reading of the obstructions recorded above: each of them is sharp for the class it sees, namely the one with n unrestricted, and the conjecture is the assertion that restricting to $n = b$ contracts the interval all the way to the tree band.

The condition $\operatorname{rank} A_k \leq b$ is therefore the condition that the occupancy of a block fluctuates as little as possible at its given mean, and the conjecture asserts that minimal marginal fluctuation propagates to the extreme confinement recorded above. That suggests the right statement is one about measures rather than matrices: for a random vector $(s_1, \ldots, s_q)$ of nonnegative integers with $\sum_k s_k = p$ almost surely, each s_k exactly $\mathrm{Binomial}(b, 1/a)$, and joint generating function homogeneous of degree p and real stable, are the roots of $\mathbb{E} \prod_k (y - as_k)$ confined to the band? Every projection family induces such a law, and the scalar family does not, failing the marginal condition.

That class is rigid, which is what makes the reformulation faithful rather than a weakening. The marginal condition is itself a rigidity statement: in the polarized picture it holds if and only if the b slots inside each block are exactly jointly independent Bernoulli$(1/a)$. Consequently, any *determinantal* law satisfying the two combinatorial conditions has a rank-p projection kernel whose diagonal $b \times b$ blocks are $\frac{1}{a} I_b$, that is, it is the Naimark kernel of a projection family; and any law given by independent balls in bins satisfying them comes from an (a, b)-biregular incidence matrix, that is, from a commuting family. So the two canonical constructions of stable laws produce nothing new. At $b = 2$ more is true: for $(p, q, a, b) = (4, 6, 3, 2)$ and $(3, 6, 4, 2)$ the affine hull of the stable laws in the class coincides with the affine hull of the projection families, of dimensions 54 and 14 against 78 and 38 for the two combinatorial conditions alone. The measure formulation is therefore not a relaxation at $b = 2$.

The two combinatorial conditions alone do not suffice, and the point at which they fail is sharp: over all laws satisfying them the largest root is confined to the band only for q below an explicit threshold, and it reaches the trivial value ab exactly at $q = a^b$, the smallest q at which the binomial profile $n_j = q\binom{b}{j}(a-1)^{b-j}a^{-b}$ is integral. Stability is what must supply the rest, and it cannot be seen by any argument that symmetrizes: the divisibility constraint that stability imposes, namely that $(w + a - 1)^{b-1}$ divides $\mathbb{E}\big[s_l\, w^{s_k}\big]$, becomes vacuous on exchangeable laws. Negative association alone is likewise insufficient, by an explicit permutation counterexample whose polynomial has a root at ab.

The size of the claim is also worth recording. Since E_1 and E_2 are universal, the root distribution of μ has mean a and variance $a(b-1)$ for every admissible family. Measured against that, the lower band edge sits between 1.35 and 1.92 standard deviations below the mean over the range $3 \le a \le 100$, tending to 2 as a grows. The conjecture is therefore a statement of extreme concentration: the least of p roots is confined to about two standard deviations, where an unstructured sample of that size would reach out to order $\sqrt{\log p}$.

12.1 The extremal problem in dimension four

Graphs are not always extremal in that class, however, and the smallest case can be settled exactly. This is no longer needed for the band — at $p = 4$, $b = 2$ the reflection gives $\mu(t+a) = t^4 - 2at^2 + \beta$, and real-rootedness alone forces $0 \le \beta \le a^2$, hence $t^2 \le 2a \le 4(a-1)$ — but it identifies the extremal families. Take $b = 2$, $p = 4$, so $q = 2a$ and $\sum_k P_k = aI_4$. Writing $g_{ij} = \operatorname{tr}(P_iP_j)$ and $h_{ij} = \operatorname{tr}\big((P_iP_j)^2\big)$,

Proposition 34. *For rank-2 orthogonal projections $P_1, \ldots, P_q$ on $\mathbb{R}^4$ with $\sum_k P_k = \frac{q}{2} I$,*

$$E_4 = \frac{q^4}{16} - \frac{q^3}{4} + \frac{q}{2} + \frac{1}{4}\sum_{i \ne j}\big(g_{ij}^2 - h_{ij}\big),$$

and every summand is nonnegative. Equality holds exactly when $\operatorname{rank}(P_iP_j) \le 1$ *for all $i \ne j$.*

Nonnegativity is immediate once the right object is named: for $i \ne j$ put $X = P_jP_iP_j \succeq 0$; then $\operatorname{tr}\big((P_iP_j)^m\big) = \operatorname{tr}(X^m)$ for every $m \ge 1$, so $g_{ij}^2 - h_{ij} = (\operatorname{tr} X)^2 - \operatorname{tr} X^2 = 2\sum_{r<s} \lambda_r\lambda_s \ge 0$. The last step is formalized in `RamaLean/TracePSD.lean`: `sq_sum_sub_sum_sq` gives the identity $\big(\sum f\big)^2 - \sum f^2 = \sum_i \sum_{j \ne i} f_i f_j$ over any finite index set and `sum_sq_le_sq_sum` the inequality under nonnegativity, with the spectral input carried as explicit hypotheses on the matrix form.

Identifying a 2-plane $U \subset \mathbb{R}^4$ with its unit simple bivector $\omega_U \in \Lambda^2\mathbb{R}^4 \cong \mathbb{R}^6$ gives $g_{ij}^2 - h_{ij} = 2\langle \omega_i, \omega_j \rangle^2$, so the Welch bound for q unit vectors in a 6-dimensional space yields

$$E_4 \;\geq\; \frac{q^4}{16} - \frac{q^3}{4} + \frac{q^2}{12},$$

with equality exactly when $\{\omega_k\}$ is a tight frame of *simple* bivectors. This gives 30, 400/3, 1150/3 and 876 at $q = 6, 8, 10, 12$, matching the values found numerically.

At $q = 6$ the bound reads $E_4 \geq 30$, attained by the incidence family of K_4; and there the tight-frame condition degenerates to "orthonormal basis", which is the one value of q at which it can be met by simple bivectors alone. That is why graphs are extremal at $q = 6$ and not at $q = 8$, where the minimum 400/3 lies strictly below the graph minimum 134: the coincidence is the degeneration, not a principle. Even at $q = 6$ the graphs are only a slice of the minimizers — the equality locus is 4-dimensional modulo $O(4)$, and taking the ω_k from the six icosahedral axes, with the two $\mathbb{R}^3$ components Galois conjugate over $\mathbb{Q}(\sqrt{5})$, produces a noncommuting family with all $g_{ij} = 4/5$ and $E_4 = 30$. A lower bound of the same shape is known in the random setting: Brito, Dumitriu and Harris [9] show that a random bipartite biregular graph satisfies $\eta^+_{\min} \geq \sqrt{d_1 - 1} - \sqrt{d_2 - 1} - \varepsilon_n$ with high probability. That is an asymptotic statement about adjacency eigenvalues of random graphs; Conjecture 10 is a deterministic statement about matching polynomial roots of every graph.

13 What a proof must contain

Each of the methods that does not reach the band fails for a reason that can be named exactly, and each such reason is a property that a proof must have. We collect them here as a specification rather than as a list of failures: a proof must not induct on vertex deletion, must not be a side-constant specialization of the multivariate matching polynomial, must control deficient vertices of the path tree, and must not pass through Floquet theory. Every one of these is sharp, in the sense that the method in question is exactly optimal for the class of objects it can see; that is the content of Section 12.

The lower bound, unlike the upper bound, is not monotone under vertex deletion. The Heilmann–Lieb argument for $|x| \leq 2\sqrt{\Delta - 1}$ inducts on vertex deletion, which is available because Δ only decreases. The quantity $|\sqrt{a-1} - \sqrt{b-1}|$ has no such monotonicity: deleting one vertex of the larger side of $K_{3,4}$ produces $K_{3,3}$, which is regular, and the gap collapses to zero — for $a = b$ one has $c = 2\sqrt{m}$, so the interval $[c - 2\sqrt{m},\, c + 2\sqrt{m}]$ has its lower end at 0 and the statement is empty. So the naive induction by vertex deletion, which is what yields the upper bound, cannot prove the lower one.

That obstruction can be strengthened, and in a way that rules out the whole family of such arguments rather than one instance of it. Every b-regular bipartite graph is an induced subgraph of some (a, b)-biregular graph, by an explicit padding construction; and over b-regular bipartite graphs the least root of ν_G tends to 0, in closed form $2\sin(\pi/2N)$ for $b = 2$. So on the class obtained by closing the biregular graphs under vertex deletion, the infimum of the least root is 0, and the inequality to be proved is simply false there. Any argument that deletes vertices and applies an inductive hypothesis — cavity recursions, stability transfer, the Heilmann–Lieb induction — is an induction over that closure, and therefore cannot work, not because biregularity is lost but because the statement itself fails. For $\mathrm{AG}(2, 3)$, where

$(s-t)^2 = 0.10102$, the minimum over all choices of d deleted vertices on the larger side runs 0.3593, 0.2424, 0.1383, 0.0549 for $d = 0, 1, 2, 3$: it drops below the target at depth three.

A second route, substituting side-constant values into the multivariate matching polynomial of Heilmann and Lieb, fails for a reason that is visible in one line. Writing $M_G(z;x) = \sum_M z^{|M|} \prod_{v \notin M} x_v$ and setting $x_v = \alpha$ on the p-side and $x_v = \beta$ on the q-side gives $\sum_k m_k z^k \alpha^{p-k} \beta^{q-k}$, in which the exponent pair contributes c^{p-q} under $(\alpha, \beta) \mapsto (c\alpha, \beta/c)$ independently of k; so the specialization depends on α and β only through the product $\alpha\beta$. Since the image of a product of two upper half-planes is $\mathbb{C} \setminus [0, \infty)$, stability under this substitution says exactly that the roots are real and nonnegative, and nothing more. In particular it says the same thing when $a = b$, where the conclusion to be proved is empty. No choice of α and β can see $a - b$.

Two obstructions remain in the direct approach, both caused by cycles in G. First, a leaf of the path tree that is an A-vertex carries $R_A = y > 0$, of the wrong sign. This one is disposed of: a maximal self-avoiding walk ending at $w \in A$ contains all a neighbours of w, hence at least a vertices of B and therefore at least a vertices of A, so

$$p < a \implies \text{the path tree has no } A\text{-leaf},$$

which covers every $K_{p,q}$ with $p < q$. Second, and this is what blocks the argument, an internal A-vertex of the path tree may have $k < P$ children, the walk having already consumed part of its neighbourhood; the bound $R_A \le y - P\alpha/(\alpha + Q)$ then fails. With $k = 1$ closure would require $\alpha(1 - \alpha - Q)/(\alpha + Q) \ge y$, impossible for $b \ge 3$ since $Q \ge 2$. Deficiency occurs already in $K_{3,4}$, where the walk $a_1 b_1 a_2 b_2 a_3$ leaves a_3 with two children rather than three. What the argument as it stands establishes is that a *full* truncated biregular tree carries no eigenvalue in the gap, consistent with the deficient subtree P_8 carrying one; that is a statement about the tree and not about G. Controlling deficient A-vertices is the open problem.

The case $b \ge 3$ of the biregular family is the remaining open one, since $b = 2$ is Theorem 11; note that every graph named below has $a \ne b$, so the statement being tested is not the empty one. It has been tested at $d = 1$ on $K_{3,4}, K_{3,5}, K_{3,6}, K_{3,7}, K_{3,8}, K_{4,5}, K_{4,6}, K_{4,7}, K_{5,6}$ and a $(3,2)$ design incidence graph, with smallest-root ratios between 1.51 and 3.33 and no root in a gap.

The natural attack is Godsil's recursion $R_v = x - \sum_u 1/R_u$, whose fixed points on the (a,b)-biregular tree are real precisely off the spectrum. Asking that the region $R_b \in [x, P/x)$, $P = a - 1$, $Q = b - 1$, be closed under the recursion reduces to $u^2 - (2P + Q)u + P^2 > 0$ at $u = x^2$, i.e. to $u < \big(2P + Q - \sqrt{Q^2 + 4PQ}\big)/2$. This is satisfied throughout the gap with room to spare: the ratio of that threshold to $(\sqrt{P} - \sqrt{Q})^2$ ranges from 2.7 to 37.7 over the cases above. So the analytic part of an induction is not the difficulty.

What blocks it is the boundary. A leaf of the path tree carries $R = x > 0$, while the argument wants $R < 0$ at a-vertices, and mixed signs among the children of a b-vertex leave the sum uncontrolled. One might hope to exploit that the counterexample to the naive subtree argument, P_8 inside the $(3,2)$-biregular tree, is deficient at *internal* vertices, whereas a path tree is full at internal vertices; but that is false for graphs of small girth, where a self-avoiding walk can be blocked early — in K_4 the walk $0, 1, 2$ admits a single extension. We therefore leave $b \ge 3$ open.

A proof of Conjecture 10 cannot go through Floquet theory. For $b_1 = b$ the deck group of the universal cover is the free group F_b, and Floquet theory is exactly the statement that for an *abelian* deck group the fibres are indexed by the characters of that group; the two coincide only at $b = 1$, where $F_1 = \mathbb{Z}$. For $b \geq 2$ the correct index set is the unitary dual of F_b, and the relevant object is the adjacency operator in the reduced C^*-algebra of F_b. This is not merely a technical obstruction: the computations of Section 3 show that the per-component weight in the exponential formula, a single graph for $b = 1$, becomes a sum over the many non-isomorphic connected ℓ-covers for $b \geq 2$.

That framework is not empty. As $d \to \infty$, independent uniform random permutations are strongly asymptotically free [22], and consequently the spectrum of a random d-lift of G converges to spec(T). Conjecture 10 is a finite-d statement about the expected characteristic polynomial rather than an asymptotic statement about typical lifts, and its content is precisely that the containment holds uniformly in d, with no exceptional set. What the $b_1 = 1$ case shows is that the interval in [4] is not the natural receptacle for the roots, and that the correct one is spectral rather than metric.

14 Related work

Two ingredients are classical and are used, not claimed. The factorization of the characteristic polynomial of a cyclic cover into twisted determinants at roots of unity is the block-circulant diagonalization [24, 25]; in the infinite-cover limit it is the Floquet–Bloch decomposition of a periodic discrete operator, and $y = -P/(2Q)$ is the corresponding Floquet discriminant. The identity $\Phi_{G,r} = \chi_G \cdot \mu_{r-1,G}$ is [4]. Closest in form is McSorley and Feinsilver [23]. For a path whose edges are cyclically labelled with t labels they establish a (τ, Δ)-recurrence $\Phi_N = \tau\Phi_{N-1} - \Delta\Phi_{N-2}$, which is the algebraic skeleton of (3), and their Theorem 4.4 solves it as $\Delta^{N/2}U_N(\tau/2\sqrt{\Delta})$, exactly the form used here. In the univariate specialization their τ is the matching polynomial of the cycle C_t and $\Delta = (-1)^t x_1 \cdots x_t$ is a monomial; passing between their counting normalization and the characteristic one used here is the homogenization $V_d \mapsto c^d V_d$, under which a monomial Δ becomes $\mu_{G-V(C)} = 1$. Their Theorems 4.4 and 4.5 together give $U_{dn+n-1} = U_d(T_n)U_{n-1}$, the identity invoked in Section 1 to pass between Theorem 1 and Hall's conjecture, so that bridge was already available in 2009. Their parameter is the number of label periods along a path and their graphs are paths, cycles and rooted trees; there is no covering and no cycle with trees attached. What is added here is the pair $(\mu_G, \mu_{G-V(C)})$, whose second entry is a genuine polynomial recording the attached trees rather than a monomial, together with the covering interpretation that makes the recurrence a statement about $\mu_{d,G}$.

15 The formalization

The core algebraic derivation — the exponential formula, the closed form of Theorem 1 for all n, r (equivalently, everything from the cycle weights (1) to the closed form), its factorization, and the parity Corollary 5 — is machine-checked in Lean 4 over Mathlib (v4.30). The main entry points:

- `Paper2.expFormula` — the exponential formula (2) in integer form $r\,A_r = \sum_k r^{(k)} f(k) A_{r-k}$ over any commutative ring, proved via Mathlib's Cauchy cycle-type count (`Equiv.Perm.card_of_cycleType_mul_eq`);

- `Paper2.thm1_Sr` — Theorem 1 in the form: the expected cycle weight (1) equals the closed form, for all n, r, over any $\mathbb{Q}$-algebra domain;

- `Paper2.cG_factored`, `Paper2.quotient_parity` — the factorization and Corollary 5.

The three remaining corollaries are formalized only in part: Corollary 2's generating function appears as the finite recurrence (2) rather than a literal power-series identity; of Corollary 3 only the underlying trigonometric factorization (`cor1_factored`) is checked, not the "all roots in $[-2, 2]$" statement end-to-end; and Corollary 4 (`cor2_phi3`) is a bounded `native_decide` over $n, r \leq 6$.

Trust base for the general theorems: `propext`, `Classical.choice`, `Quot.sound` only (checked with `#print axioms`); the bounded `native_decide` checks additionally invoke `Lean.ofReduceBool`. No `sorry`. The single remaining unformalized ingredient of the *graph-theoretic* statement is the sentence before (1): a lift of a cycle is a disjoint union of cycles — elementary covering-space combinatorics for which Mathlib currently lacks the vocabulary.

For the biregular half, `RamaLean/BipartiteMatchingPoly.lean` supplies a matching polynomial for bipartite graphs, Mathlib having none, and formalizes: the deletion recursion $Q_G = Q_{G-v} - z \sum_{u \sim v} Q_{G-v-u}$ (`bipQ_delete_left`); the general Taylor shift $[X^j] f(X + c) = \sum_i \binom{i+j}{j} f_{i+j} c^i$ (`taylor_coeff_of_natDegree_le`), from which the coefficient formulas at $p-1$, $p-2$ and $p-4$ follow; the count $2m_2 + |\text{confL}| + |\text{confR}| = |E|(|E| - 1)$ with the two conflict classes as degree sums; and the combinatorial inputs to Theorems 13 and 14 — that three pairwise-meeting edges of a bipartite graph share a vertex (`three_meeting_share`), that the conflict degree of an edge is $(d_u - 1) + (d_v - 1)$ (`conflict_degree`), the trichotomy for four-cycles of the conflict graph (`four_cycle_trichotomy`), and the identity the whole expansion runs on, that the matching indicator is the alternating sum over subsets of the conflicting pairs (`alt_sum_conflictPairs`). Several of these, including the trichotomy, depend on no axioms at all. `RamaLean/Paper2Unicyclic.lean` further formalizes that the lower band edge is $(\sqrt{P} - \sqrt{Q})^2$ and so vanishes exactly when $P = Q$ (`gap_eq_zero_iff`), which is the triviality recorded above.

The inclusion–exclusion itself is formalized as a chain. The matching indicator is the alternating sum over subsets of the conflicting pairs of a set of edges (`alt_sum_conflictPairs`); substituting it turns m_k into a double sum (`mCount_eq_sum_alt`); because conflict is intrinsic to a pair of edges, the inner index set depends on the outer one only through a containment (`powerset_conflictPairs_eq`), so the two summations exchange (`mCount_inclusion_exclusion`); the resulting weight is a binomial coefficient, by a bijection between the k-subsets containing a fixed V and the $(k - |V|)$-subsets of the complement (`card_powersetCard_filter_superset`), giving

$$m_k = \sum_S (-1)^{|S|} \binom{|E| - |V(S)|}{k - |V(S)|}$$

over sets S of conflicting pairs (`mCount_closed_form`); and since the weight sees S only through $|S|$ and $|V(S)|$, the sum fibers over that pair (`mCount_grouped`, `mCount_grouped_range`), which is the grouping by subgraph type. The chain is exercised end to end at $k = 1$ (`mCount_one_via_inclusion_exclusion`): every nonempty set of conflicting pairs has at least two endpoints, so only the empty set survives the binomial and contributes $\binom{|E|}{1}$, reproving $m_1 = |E|$ through the inclusion–exclusion rather than by the direct bijection, so that two independent routes cross-check each link.

What is *not* formalized is the evaluation of the ten resulting counts as explicit functions of (p, a, b) and $C_4(G)$ — the step at which `three_meeting_share` and `four_cycle_trichotomy` are applied — together with the numerical verifications. One obstacle there is worth recording: the formalized conflict set consists of *ordered* pairs, so each edge of the conflict graph occurs twice. This leaves the inclusion–exclusion untouched, since the alternating sum vanishes over any nonempty index set whether or not it double counts, but it means the grouping enumerates more configurations than the ten unordered types above, and aligning the two would require rebuilding the conflict set on unordered pairs.

The operator side of Sections 4–12 is formalized in four further files, all on the standard three axioms and with no `sorry`. `RamaLean/SignReflection.lean` carries Proposition 32 to the point where only the rank-one expansion of μ is left by hand: `sign_avg_reflect` proves that $\sum_{\varepsilon\in\{\pm1\}^q} \det(tI - \sum_k \varepsilon_k S_k)$ is even or odd in t according to the parity of the dimension, with *no* hypothesis on the S_k, by the involution $\varepsilon \mapsto -\varepsilon$; `sum_prod_sgn` is character orthogonality on the Boolean cube, proved by the same involution applied to one coordinate; `sum_multiaffine` deduces that averaging a multi-affine function over signs returns 2^q times its constant term, which is exactly what $\prod_k (1 - \partial_{z_k})$ extracts; and `mixedChar_reflect` and `band_edges_swap` supply the shift and the identity $2a - (\sqrt{a-1}+1)^2 = (\sqrt{a-1}-1)^2$ that exchanges the band edges. `RamaLean/ProductBound.lean` proves Proposition 19 (`no_root_above`: a convex combination of products of positive numbers is positive) together with the exact overshoot $ab - (s+t)^2 = (\sqrt{m}-1)^2$ (`edge_gap`) and its vanishing exactly at $m = 1$ (`edge_gap_eq_zero_iff`), which is the case $a = b = 2$ where the band is proved outright; it also proves Proposition 20, that at $b = 2$ the two band edges are the single condition $\pi_{\max} \le 4(a-1)/a^2$ (`band_iff_pi`), and that the strictly weaker $\pi_{\max} \le 4/a$ already yields $a + 2\sqrt{a}$, which beats the Marchenko–Pastur edge for every a and the product bound ab for $a > 4$ (`weak_bound`, `weak_beats_mp`, `weak_beats_ab`). `RamaLean/GramDet.lean` discharges the exterior-algebra inputs by writing them in Gram coordinates, where each becomes a determinant statement. `det_border` is the bordered determinant formula $\det\left[\begin{smallmatrix} a & r^{\mathsf{T}} \\ c & G \end{smallmatrix}\right] = a \det G - r^{\mathsf{T}} \operatorname{adj}(G) c$, which is exactly (14); `det_gram_eq_zero_of_dep` is $f_k \wedge \omega'_k = 0$, the vanishing of a Gram determinant on a dependent family; and `hsimple_of_border_zero` combines them into $\|f_k\|^2 \|\omega'_k\|^2 = \|\iota_{f_k} \omega'_k\|^2$, which is precisely the hypothesis `hsimple` used below — so that hypothesis is now proved rather than assumed. `RamaLean/Tightness.lean` supplies the other input, `htight`: that $\operatorname{Adj}(A) = aI$ forces $\sum_k \iota_{f_k} \omega'_k = 0$. Working in coordinates, with a rank-two block presented by its two spanning vectors, the polarization step becomes three elementary facts — `iota_antisymm`, that $\langle a, \iota_b \omega\rangle = -\langle b, \iota_a \omega\rangle$, which is a two-term `ring` identity; `iota_proj_of_orthogonal`, that compressing the block to $e^\perp$ does not change the pairing against f_k; and that each $\iota_{f_k} \omega'_k$ already lies in $e^\perp$. `tight_sum_contraction_eq_zero` assembles them, and `crossTerm_nonneg_of_tight` composes the result with `CrossTerm.crossTerm_nonneg`, so that Theorem 24 is machine-checked from the tightness hypothesis alone. `RamaLean/Plucker.lean` closes the last link, the identification of `CrossTerm`'s coordinate vector `w` with the Plücker coordinates of ω'_k. Representing $b_1 \wedge b_2$ by its antisymmetric array on *ordered* pairs scaled by $1/\sqrt{2}$ makes `pl_dot`, that the resulting inner product is the Gram determinant, hold with no $i < j$ bookkeeping: the double count and the two factors of $1/\sqrt{2}$ cancel. `iota_eq_comb` writes $f_k = \iota_e \omega_k$ in the compressed plane with explicit coefficients $-\langle e, b_2\rangle$ and $\langle e, b_1\rangle$, and `hsimple_of_comb` then makes `hsimple` a `ring` identity rather than an assumption. `crossTerm_nonneg_plucker` assembles everything: **for a tight family of rank-two blocks and any unit vector, $C_2 \geq 0$ is machine-checked with no hypothesis remaining**.

The same holds at every level. `GramDet.border_gram` identifies the bordered Gram matrix of $[v \mid M]$ with the Gram matrix of the bordered family; `border_det_eq_zero_of_mem_span` then gives $f \wedge \omega = 0$ whenever f lies in the column span, which it always does; and `hsimple_of_mem_span` combines them into $\|f\|^2\|\omega\|^2 = \|\iota_f \omega\|^2$ for a wedge of *any* number of vectors. So the hypothesis `hsimple` of Theorem 25 is discharged at every r, not only at $r = 2$, and what Theorem 25 asserts — that each S-summand of C_r is a Schur square minus the square of a contraction — is machine-checked unconditionally. Finally `gram_det_basis_of_injective` and `gram_det_basis_of_not_injective` prove that for a coordinate family the Gram determinant of a set of blocks is 1 on matchings and 0 otherwise, that is $F_A = \mu_G$; that is the first of the two inputs to Proposition 26, and the second, the Kesten–McKay limit, is the only analytic ingredient anywhere in this section and is cited rather than proved. `RamaLean/CrossTerm.lean` carries Theorems 24 and 25. `crossTerm_eq_sq_sub` is the general level: with no hypothesis beyond `hsimple` it proves $\sum_{k \neq l}[\langle f_k, f_l\rangle\langle w_k, w_l\rangle - \langle u_k, u_l\rangle] = \|\sum_k f_k \otimes w_k\|^2 - \|\sum_k u_k\|^2$, which is the S-summand of Theorem 25 once $w_k = \omega'_k \wedge \omega'_S$ and $u_k = \iota_{f_k}(\omega'_k \wedge \omega'_S)$. Its two geometric inputs are hypotheses — `hsimple`, that $f_k \wedge \omega'_k = 0$ in its metric form $\|f_k\|^2\|\omega'_k\|^2 = \|u_k\|^2$, and `htight`, that $\sum_k u_k = 0$ — and `crossTerm_eq_sq` specializes it to $r = 2$, where tightness kills the second square, and proves that they force C_2 to equal $\sum_{i,j}\big(\sum_k (f_k)_i(\omega'_k)_j\big)^2$, with `crossTerm_nonneg` and `crossTerm_eq_zero_iff` the consequences. The supporting `sum_gram_prod_eq_sum_sq` is the Schur product theorem in the case needed, that $\sum_{k,l}\langle f_k, f_l\rangle\langle w_k, w_l\rangle$ is a sum of squares. No exterior algebra is involved: the bivectors enter only through their coordinates. `RamaLean/VertexSplit.lean` carries the Gram form of Theorem 22. Its core, `det_add_vecMulVec`, is the adjugate form (13) of the matrix determinant lemma; Mathlib has the lemma only as $\det(A + uv^\top) = \det A \cdot \det(1 + v^\top A^{-1}u)$ and records the adjugate form as an explicit `TODO`, so this supplies it, under the hypothesis that $\det A$ is a unit — which is what the application needs, the Gram matrix of an independent family being positive definite. `det_le_det_add_vecMulVec` is the inequality the recursion consumes, $M_r(A) \geq M_r(A^{(e)})$, deduced from `AdjugatePSD.adjugate_posDef`. The singular case is scoped out as in `AdjugatePSD`: both sides of (13) are polynomial in the entries, so it extends by continuity from $A + \varepsilon I$, and that limit is not formalized. `RamaLean/CavityThreshold.lean` carries Proposition 23: `cavity_step` is the induction step, that child ratios at least $x/2$, nonnegative weights of total at most a, a nonpositive remainder and $4a \leq x^2$ force the parent ratio to be at least $x/2$ again; `threshold_iff` is its sharpness, that $x - 2a/x \geq x/2$ holds *if and only if* $x \geq 2\sqrt{a}$; `double_root_at_threshold` is the structural form of the same statement, the factorization $\rho^2 - 2\sqrt{a}\,\rho + a = (\rho - \sqrt{a})^2$ that collapses the interval of admissible thresholds to a point; and `no_slack_propagates` is Proposition 28; `cavity_step_deficient` is Proposition 30, the strengthened step; and `compression_sum`, `trace_le_of_rayleigh` and `remainder_upper` are Proposition 31. Proposition 29 is `Tightness.adj_compressed`, with the per-block splitting `Tightness.iota_split`. `RamaLean/KestenMcKay.lean` carries Proposition 26: `radical_at_edge` that the radical is exactly 2 at the threshold, `G_at_edge` the closed form there, `inv_G_div_edge` the constant $1 + 1/(1 + \sqrt{a})$, and `excess_eq`, `excess_pos`, `excess_lt` that the excess is exactly $1/(1+\sqrt{a})$, positive for every $a > 1$ and below any ε once $\sqrt{a} > 1/\varepsilon$. The two inputs it does not formalize are the ones that are not arithmetic: that F_A is the matching polynomial, which is Theorem 21, and the Kesten–McKay limit, which is cited. The vertex recursion of Theorem 22 and the sign of X_e enter as hypotheses; what is machine-checked is that they close the induction, and that they close it exactly at $2\sqrt{a}$. `RamaLean/TracePSD.lean` and `RamaLean/AdjugatePSD.lean` carry the two positivity facts the extremal computation and the kernel recursion run on. In both, the standard input is a visible hypothesis rather than a hidden one: `trace_sq_le_sq_trace` takes the spectral decomposition of a positive semidefinite matrix as an explicit assumption, and `adjugate_posDef` is proved in the definite case, the semidefinite limit $A + \varepsilon I$ being left by hand.

Remark 35. Everything here is stated for the uniform permutation model; by [4] the quotient coincides with the $(r-1)$-matching polynomial, so Theorem 1 can alternatively be deduced from [5]. We believe the generating function of Corollary 2, the evaluation of Corollary 4, and the formalization are new.

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